Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Supplementary Material

Sections S1 to S5 specify the optical model, training, reconstruction and evaluation. Full results, structural tests and acquisition controls follow in Secs. S6 to S8; Secs. S9 and S10 give complementary uncertainty analyses and neuronal display conventions. Section, equation and figure numbers without an S prefix refer to the main paper. Unless stated otherwise, additional experiments use the 12 prespecified neuronal crops, with five direction subsets at $C > 1$. Table S1 summarizes the evaluation units and controls.

S1. Optical implementation and discretization

Coordinates and angular scan. The optical axis is z and $\mathbf{r} = (x, y)$ is transverse position. In the simulated input medium, $n = 1$ and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$. The incident direction is

$$\mathbf{k}_t/k_{\mathrm{ex}} = (\cos\theta_t \cos\alpha_t, \cos\theta_t \sin\alpha_t, \sin\theta_t). \tag{S1}$$

Thus, θ is the implemented vertical-plane parameter, not a polar angle from the optical axis. We use $\alpha_m = m\pi/16$, $m = 0, \ldots, 15$, with eight equally spaced θ values in each of $[\pi/6, \pi/3]$ and $[2\pi/3, 5\pi/6]$. The second interval changes the sign of the transverse component while retaining positive longitudinal propagation. The resulting 256 directions are distinct. For $C = 4$ and 8, five fixed random subsets of 64 and 32 directions are shared across encoders.

Propagation and sampling. The incident field is uniformly x-polarized before phase modulation. Angular-spectrum propagation over $D = 6.6\,\mu\mathrm{m}$ carries the field from the plate to the intermediate illumination image plane. A finite NA-0.8 pupil filters the modulated angular spectrum before a vectorial Richards–Wolf relay generates three specimen-field components [8, 10]. Incident direction and transmitted angular spectrum are different quantities; the input direction range need not lie entirely inside the relay pupil. We sum the component intensities, modeling isotropically averaged fluorophore orientations. The phase plate is an ideal thin element; its feature pitch and propagation distance are specimen-equivalent. Detection uses an NA-0.1 emission intensity PSF. Excitation and emission share 532 nm in this simulation; a physical implementation would use the actual spectral bands. No scattering, aberration or fluorescence saturation is modeled.

The specimen field is sampled laterally at $0.08\,\mu\mathrm{m}$ and axially at $0.32\,\mu\mathrm{m}$. Lateral binning gives an isotropic $0.32\,\mu\mathrm{m}$ analysis grid of $32 \times 96 \times 96$ cells. The reconstruction basis averages $2 \times 2 \times 2$ analysis cells into each $0.64\,\mu\mathrm{m}$ voxel. Camera pixels average 8×8 analysis samples laterally, giving $2.56\,\mu\mathrm{m}$ pixel pitch and 12×12 samples per image. Distances and pitches are expressed in specimen-equivalent coordinates.

Image formation. Let $X_0(\mathbf{r}, z)$ denote fluorescence response per unit excitation and h_{em} the emission intensity PSF. With linear fluorescence and laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r} - \mathbf{r}', z_{\mathrm{focal}} - z') \times X_0(\mathbf{r}', z') I_t(\mathbf{r}', z')\, \mathrm{d}^2\mathbf{r}'\, \mathrm{d}z'. \tag{S2}$$

Camera pixel p, occupying D_p, records $\bar{y}_{t,p} = |D_p|^{-1} \int_{D_p} g_t(\mathbf{r})\, \mathrm{d}^2\mathbf{r}$. Emission adds incoherently over the specimen before this average. Excitation multiplies the specimen before detection blur; the detector need not resolve the illumination pattern itself.

Aperture and control encoders. The phase plate has $0.32\,\mu\mathrm{m}$ features and initially independent phases uniform on $[0, 2\pi)$. It is simulated on an aperture twice the specimen width, then the excitation is cropped to the specimen. Only the central plate region is optimized; its independent random surround remains fixed. The defocused amplitude controls use binary masks with the same feature pitch and 50 % open pixels at the same plate plane. The in-focus DMD uses independently drawn binary patterns relayed into focus. These are controls for the tested fixed-focus acquisition, not a comparison against all DMD microscopy configurations.

A finite illumination edge can itself encode depth when tilted. The plate-free control therefore uses an aperture four times the specimen width. Its residual central/border angular modulation is about 1.0/1.2 %, compared with 5.6/37 % at twice the width. The $4\times$-aperture plate-free study is kept separate from the $2\times$ phase/mask comparison. The untrained plate changes little across these apertures (centroid error 0.446 versus $0.431\,\mu\mathrm{m}$ at $C = 8$, pretrained PnP), but that does not make their learned-versus-plate-free contrast a single-aperture experiment.

Integrated operator. The detection footprint $g_{p,i}$ includes the emission PSF, camera-pixel average and fine-cell integration weight. Reusing it in $A_{(t,p),j} = \sum_{i \in \mathcal{P}_j} I_t(\mathbf{r}_i, z_i) g_{p,i}$

Study	Reference units	Draws	Results
Five designs, four methods	60 crops, six cells, three animals	5	Tables S2, S3 and S6
Optical sampling seed 1	Same 60 crops	5	Tables S4, S5 and S9
Phase/amplitude controls	12 crops, pretrained PnP	5	Table S11; centroid/intensity metrics
In-focus DMD	12 crops, three iterative methods	5	Section S6
Point-emitter diagnostics	25 positions, 16 depths	–	Section S6
Full scan ($C = 1$)	12 prespecified crops	1	Tables S15 and S17
Compression, dose, alignment	12 prespecified crops	5	Tables S15, S20 and S21
Noise and reduced-dose crops	12 prespecified crops	5	Tables S18 and S19
Merged-depth alternatives	59 eligible test crops	5	Main Fig. 4; Sec. S7
Equal-branch crossings	36 phantoms, including zero gap	5	Tables S12 and S13
Weak/three-filament tests	Cases specified in Sec. S7	5	Table S12
Reduced-dose crossings	24 phantoms, including zero gap	3	Table S19 and Fig. S4
Conformal intervals	35 calibration crops; 12 test crops	–	Table S22
Reduced-field bounds	24 phantoms; 48 neuronal windows	3	Section S9

Table S1. Evidence map. Draws are averaged within crops or phantom cases before paired statistics. Crops from the same cell are grouped during the cell bootstrap. The reduced-field row counts noise realizations per phantom. Conformal calibration follows its separate protocol in Sec. S9. Both are complementary analyses.

avoids propagating each reconstruction basis element separately. Let B denote the fine-grid measurement operator and U the operator that replicates each reconstruction coefficient over its constituent fine-grid cells. Equality $\mathsf{A} = \mathsf{BU}$ is an algebraic statement for the selected piecewise-constant basis. It does not assert that a real specimen is constant within a voxel. For a fine-grid specimen $\mathbf{x}_f$, the voxel-averaged reference is $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, where R averages within reconstruction voxels. Normalized noisy measurements satisfy

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{UR}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{S3}$$

The retained residual is caused by specimen structure outside the reconstruction basis, not by an algebraic discrepancy between $\mathsf{A}\mathbf{x}$ and $\mathsf{BU}\mathbf{x}$.

Computational scope. After detection-footprint precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF \log F)$ for constructing columns through separate FFT-based forward simulations: T illumination patterns, P camera samples, F fine-grid cells and N reconstruction voxels. Representative timings on an RTX 4500 Ada are approximately 20 minutes for 1000 optical-training steps, six minutes to assemble a full plate operator, 4–6 seconds for a PnP reconstruction at $C = 8$, and 17 seconds for TV.

S2. Plate training and objective comparisons

Signatures and sampling. For each lateral position and depth, we collect the detector-integrated response of its reconstruction voxel over the sampled illumination directions and camera pixels. Four directions and 24 interior lateral positions are sampled per Adam step. Each full-field depth

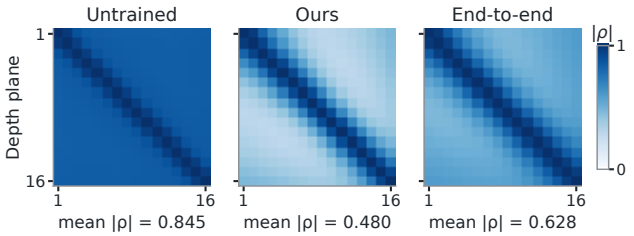


Figure S1. Absolute correlations of centered, unit-norm depth signatures for the untrained, our and end-to-end plates. Each signature uses the 32 directions of draw 0; matrix entries average over 44×44 interior positions. Values below the matrices are means of their off-diagonal entries.

signature therefore has 4×144 entries during training. Its mean over all angle–pixel entries is removed before unit-norm normalization. This is not per-pixel temporal centering. The main paper's evaluation statistic uses the same centering definition over the 32 directions of draw 0 and the 44×44 interior lateral positions (indices 2–45). It averages absolute off-diagonal depth correlations, not the squared weighted training loss. Figure S1 compares the three plates.

With zero-based reconstruction-plane indices $i, j \in \{0, \ldots, 15\}$, off-diagonal weights are

$$w_{ij} = \begin{cases} 2, & i + j = 14, \\ 1, & i + j \neq 14, \end{cases} \qquad i \neq j. \tag{S4}$$

This emphasizes reflected pairs around grid plane 7. It is an implemented weighting choice, not a universal symmetry of a phase screen or a statement that plane 7 is exactly the physical focus. Both orderings of each pair are included and the sum is divided by the total weight. Norm denominators

include 10^{-12} for numerical stability.

We use 1000 Adam steps with learning rate 0.05 and retain the final iterate, without checkpoint selection. The two training seeds change the sampled positions and directions, starting from the same phase realization and frozen surround. No specimen examples, reconstruction loss or denoiser enter this process. Normalizing a signature removes uniform power scaling from the loss; dose normalization is applied separately when generating evaluation measurements.

Local Gaussian interpretation. At one lateral position, let Q collect Z unit-norm depth signatures and $\mathsf{G} = \mathsf{Q}^\top\mathsf{Q}$. Then $\operatorname{tr}\mathsf{G} = Z$ and the unweighted off-diagonal sum is $\|\mathsf{G}-\mathsf{I}\|_F^2$, the frame potential minus its diagonal contribution Z [3]. Consider the idealized observation $\mathbf{y} = \mathsf{Q}\mathbf{x} + \boldsymbol{\epsilon}$, with independent $\mathbf{x} \sim \mathcal{N}(0, \sigma_x^2\mathsf{I})$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_e^2\mathsf{I})$. Writing $\gamma = \sigma_x^2/\sigma_e^2$, the mutual information and linear-MMSE are

$$I(\mathbf{x};\mathbf{y}) = \tfrac{1}{2}\log\det(\mathsf{I} + \gamma\mathsf{G}), \tag{S5}$$

$$\mathrm{MSE}_{\mathrm{lin}} = \sigma_x^2 \operatorname{tr}(\mathsf{I} + \gamma\mathsf{G})^{-1}. \tag{S6}$$

At fixed trace, concavity of the log and convexity of the inverse favor equal eigenvalues, giving the orthogonal optimum $\mathsf{G} = \mathsf{I}$ when attainable. Set $\mathsf{E} = \mathsf{G} - \mathsf{I}$. Since $\operatorname{tr}\mathsf{E} = 0$, the expansions near this point are

$$I = \tfrac{Z}{2}\log(1+\gamma) - \tfrac{1}{4}\left(\frac{\gamma}{1+\gamma}\right)^2 \|\mathsf{E}\|_F^2 + O(\|\mathsf{E}\|_F^3), \tag{S7}$$

$$\mathrm{MSE}_{\mathrm{lin}} = \sigma_x^2\left[\frac{Z}{1+\gamma} + \frac{\gamma^2}{(1+\gamma)^3}\|\mathsf{E}\|_F^2 + O(\|\mathsf{E}\|_F^3)\right]. \tag{S8}$$

This motivates decorrelation of axial responses. It does not derive the implemented weights, centering or normalization from the full Poisson specimen model. The actual learned signatures remain correlated, other lateral columns interact, and the full reconstruction is not a Gaussian linear-MMSE estimator. The calculation is a local interpretation of a known Gram-matrix design principle [4, 5], not a transfer guarantee or an error bound for neuronal W1.

Why angular modulation matters. Consider a nonnegative temporal response $\mathbf{e}_z$ at one camera pixel, with mean $\mu_z > 0$. Write $\mathbf{e}_z = \mu_z\mathbf{1} + \widetilde{\mathbf{e}}_z$, and define angular contrast $\kappa_z = \|\widetilde{\mathbf{e}}_z\|/(\sqrt{T}\mu_z)$. If ρ is the correlation of the mean-removed fluctuations, the raw cosine is exactly

$$c_{zz'} = \frac{1 + \rho\kappa_z\kappa_{z'}}{\sqrt{(1+\kappa_z^2)(1+\kappa_{z'}^2)}}. \tag{S9}$$

The shared mean dominates when contrasts are small, even if the fluctuations are weakly correlated. Increasing contrast and placing strong responses at different angles can reduce this overlap. At the preset field center, median contrast is 0.58/1.53 and mean raw cosine 0.83/0.41 for untrained/our plates. Only one preset position is used for this illustrative decomposition; it is not a field-wide causal ablation.

A common-footprint approximation explains why centering the full angle–pixel signature does not necessarily remove the shared angular component. If $\mathbf{q}_z = \mathbf{e}_z \otimes \mathbf{g}$ with a fixed P-pixel footprint, global centering gives correlation

$$\frac{1 - \eta + \rho\kappa_z\kappa_{z'}}{\sqrt{(1-\eta+\kappa_z^2)(1-\eta+\kappa_{z'}^2)}}, \quad \eta = \frac{(\mathbf{1}^\top\mathbf{g})^2}{P\|\mathbf{g}\|^2}. \tag{S10}$$

For a localized footprint, η can be small, so a common angular component remains. The actual operator includes depth-varying footprints and within-voxel variation; neither training nor evaluation uses this separable approximation. This provides an interpretation of the observed code, not a proof that contrast alone causes the reconstruction gain or that the objective is universally optimal.

In the saved central xz and yz excitation sections over all 256 directions, the 99.9th-percentile intensity divided by the full-volume scan mean rises from 5.5 for the untrained plate to 17.6 for our plate. The sampled maxima are 11.5 and 50.8. These section statistics describe excitation concentration, not an experimentally established saturation or bleaching threshold. The linear fluorescence model assumes the chosen operating power keeps the response unsaturated.

Worst-case coherence surrogate. Let ρ_{ij} be the inner product of two centered, unit-norm depth signatures. The comparison uses

$$\mathcal{L}_{\mathrm{coh}} = \mathbb{E}_{\mathbf{r},\mathcal{T}}\left[\frac{1}{50}\log\sum_{i\neq j}\exp(50|\rho_{ij}|)\right]. \tag{S11}$$

This smooth maximum targets the worst pair rather than the weighted average of squared correlations. It is motivated by coherence-based sensing and PSF design [4, 5, 14]; it does not implement Miniscope3D's full optical design or lateral-shift cross-coherence objective.

Depth Fisher-information surrogate. Let R contain the raw signatures in a sampled batch. The implementation uses $s = R/\operatorname{stopgrad}(\max(\operatorname{mean} R, 10^{-30})) + 10^{-3}$. Finite differences s'_z are taken over depth-plane index, centrally inside the grid and one-sided at its ends. With brightness treated as a nuisance parameter, the depth-information surrogate is

$$\mathcal{I}_z = \sum_m \frac{(s'_{z,m})^2}{s_{z,m}} - \frac{(\sum_m s'_{z,m})^2}{\sum_m s_{z,m}}, \tag{S12}$$

where m indexes angle–pixel entries. Training minimizes $-\mathrm{mean}_z \log \max(\mathcal{I}_z, 10^{-12})$. This uses the Schur-complement form of Poisson Fisher information [11], with the stated normalization and discretization. It is an implemented surrogate, not a comparison with every Fisher-based design.

Matched budget and centering ablation. All objectives use the same initial phase, sampled positions/directions, optimizer, learning rate and step count. Reconstruction parameters are validated independently for each resulting operator. These conditions match computational budgets, not convergence rates or optimal hyperparameters for every criterion. The uncentered variant omits mean removal and retains the weights and squared-correlation form. On the 12 prespecified crops with neuron-trained PnP at $C = 8$, it is essentially tied with the centered version (W1 0.207 versus $0.206\,\mu\mathrm{m}$). Centering is therefore not an established explanation of the gain over the tested alternatives.

Recorded optimization trajectories. Figure S2 shows both sampling seeds for each criterion. Each trace contains 41 values: step 1 and every 25th step through 1000, evaluated on that step's four-angle, 24-position minibatch before its update. Objective values are not comparable between criteria. The common correlation statistic centers and normalizes each sampled angle–pixel signature before averaging absolute off-diagonal correlations. It differs from the main paper's 32-angle full-field statistic. For seed 0, first-100 to last-100-step window means change from 0.741 to 0.523 for our loss, -3.490 to -4.082 for depth Fisher, and 1.062 to 1.055 for worst-case coherence; seed 1 has the same trend. Every logged batch retains at least one depth pair with absolute correlation above 0.998. That global batch maximum is distinct from the coherence loss, which averages a smooth maximum at each position. No held-out or full-operator objective trajectory was recorded. These noisy traces document progress within the shared budget; they do not establish convergence of any criterion. The comparison uses a fixed budget rather than criterion-specific stopping or learning-rate tuning.

Complete 60-crop objective comparison. The primary five-design table uses all 60 test crops and five draws per crop. The two alternative criteria are evaluated with TV, pretrained PnP, neuron-trained PnP and separately fine-tuned feed-forward decoders. Table S2 reports their full metrics, and Tab. S3 gives paired W1 changes relative to our sampling-seed-0 plate. The corresponding untrained, our and end-to-end rows are in Tab. S6. These tables use complete configurations only. Table S4 repeats the TV/PnP comparison with sampling seed 1 on the same 60 crops. The recorded comparison uses our seed-0 plate as the fixed reference for both criterion seeds; all paired W1 intervals exclude zero. As a sensitivity check, Tab. S5 compares each seed-1 criterion with our seed-1 plate and gives the same conclusion.

S3. Joint optical–decoder training

Shared optics. The jointly trained counterpart uses the same phase parameterization, initial phase, frozen random surround, vectorial relay, detector footprints and reconstruction basis as the model-only designs. The measurement matrix is assembled differentiably from the excitation and fixed detection footprints. During joint training, each sampled 32-direction excitation is normalized by its own specimen-field mean. Evaluation instead uses the full-scan dose normalization shared by every encoder. This difference is retained explicitly in the training protocol; model-only correlation losses are invariant to a uniform scale.

Regularized solve and network. For $M = 32 \times 144 = 4608$ measurements and $N = 36{,}864$ coefficients, the decoder computes

$$\begin{aligned} \mathbf{z}_\lambda &= \mathbf{m} + \mathsf{A}^\top(\mathsf{A}\mathsf{A}^\top + \lambda \mathsf{I})^{-1}(\mathbf{y} - \mathsf{A}\mathbf{m}), \\ \hat{\mathbf{x}} &= \mathcal{N}_\theta(\mathbf{z}_\lambda). \end{aligned} \tag{S13}$$

Here $\mathbf{m}$ is a constant volume filled with the training set's mean intensity. A learned log-parameter sets $\lambda > 0$, initialized from mean training noise variance divided by training voxel variance. The solve is performed in measurement space. The Gram matrix is formed in float32 with TF32 disabled, and its Cholesky factorization and triangular solve use float64. The result is projected back in float32. The map $\mathcal{N}_\theta$ is the residual 3D U-Net described below.

The network has three encoder widths, 16, 32 and 64. Each block contains two $3 \times 3 \times 3$ convolutions with batch normalization and ReLU. Two $2 \times 2 \times 2$ max-pooling stages reduce the spatial grid. Transposed convolutions of kernel and stride two upsample; lateral skip features are concatenated before the corresponding convolution blocks. A final $1 \times 1 \times 1$ convolution predicts a residual that is subtracted from the network input. The output has no nonnegative or box activation.

Training data and noise. Training uses the same 459 neuronal crops from stacks 8 and 9 of the fourth animal as the denoiser study. Stack 10 contributes 240 held-out crops, of which every fifth crop gives 48 validation volumes. Each validation uses two fixed direction subsets and fixed noise, shared by all optical designs. Test animals never enter these stages. Training augments volumes with independent flips along each axis and lateral transposition.

Supervised training generates measurements from the coarse reconstruction-basis model $\bar{\mathbf{y}} = \mathsf{A}\mathbf{x}$. The noise ap-

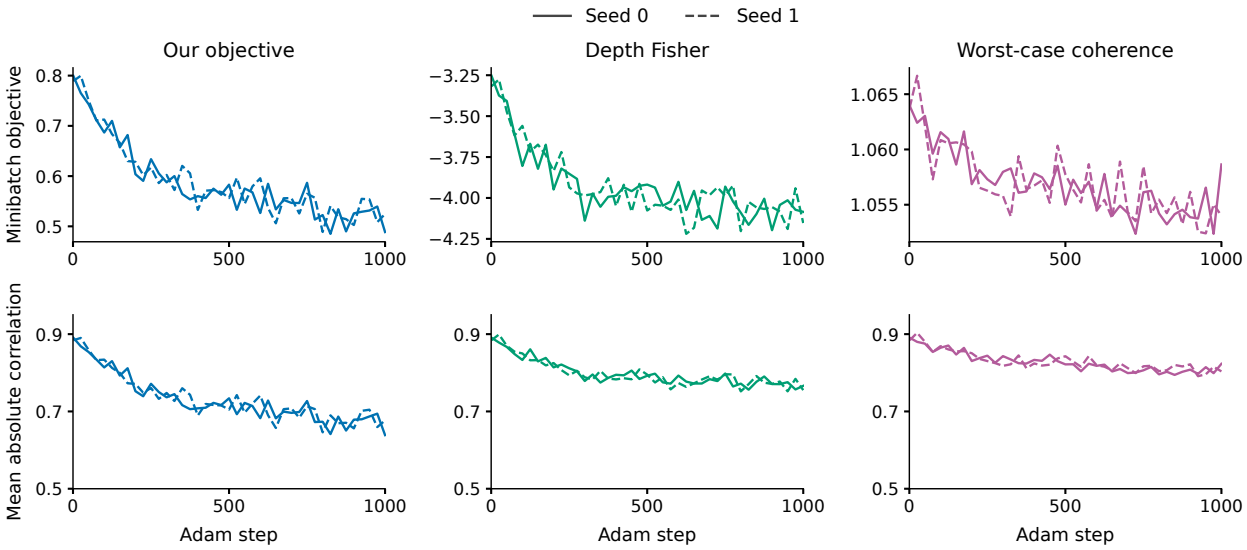


Figure S2. Recorded model-only optimization histories. Top: each criterion's own minibatch objective, with separate vertical scales. Bottom: the same mean absolute depth-correlation statistic for each criterion. Solid and dashed lines denote the two sampling seeds from the same initial phase. Values use changing minibatches and are not full-operator convergence measurements.

Reconstruction	Plate	W1 (μm)	Centroid (μm)	PSNR	FG-SSIM
TV	Depth Fisher	0.251	0.175	38.33	0.908
TV	Worst-case coherence	0.264	0.185	37.99	0.898
Pretrained PnP	Depth Fisher	0.367	0.298	37.09	0.882
Pretrained PnP	Worst-case coherence	0.390	0.320	36.85	0.871
Neuron-trained PnP	Depth Fisher	0.218	0.157	39.35	0.927
Neuron-trained PnP	Worst-case coherence	0.223	0.160	39.19	0.924
Feed-forward	Depth Fisher	0.314	0.234	37.40	0.896
Feed-forward	Worst-case coherence	0.314	0.238	37.31	0.892

Table S2. Alternative model-only objectives on all 60 crops, $C = 8$, sampling seed 0. Each row averages five draws per crop and uses the same reconstruction for all four metrics. W1 and centroid errors are in μm.

Reconstruction	Ours vs. depth Fisher	Ours vs. worst-case coherence
TV	-13.1% [-14.4, -11.5]	-17.5% [-18.8, -16.3]
Pretrained PnP	-20.9% [-22.8, -18.9]	-25.7% [-27.8, -23.4]
Neuron-trained PnP	-10.2% [-10.8, -9.3]	-12.3% [-12.8, -11.6]
Feed-forward	-15.4% [-17.0, -14.0]	-15.4% [-16.8, -14.2]

Table S3. Relative W1 changes and 95 % cell-bootstrap intervals for the primary 60-crop criterion comparison. Negative changes favor our plate. Each criterion shares the initial phase and model-only optimization budget.

proximation is

$$\widetilde{y}_i = \bar{y}_i + \sqrt{\max(\bar{y}_i, 0)/\kappa_k + 10^{-12}}\,\xi_i, \quad \xi_i \sim \mathcal{N}(0,1), \tag{S14}$$

with $\kappa_k = \kappa_0\, 256/32$ and per-crop exposure calibrated using the untrained plate. Test measurements use the common fine-grid specimen model and exact Poisson sampling. This preserves the sub-voxel specimen residual during evaluation rather than adding it to supervised training targets.

Optimization schedule. All supervised stages use batch size 16 and reconstruction MSE. The initial decoder is warmed up for 5000 Adam steps on the frozen untrained

Comparator	Reconstruction	Other W1	Our W1	Change [95% CI]
Depth Fisher	TV	0.253	0.218	-13.8% [-15.3, -12.4]
Depth Fisher	Pretrained PnP	0.370	0.290	-21.7% [-24.4, -18.8]
Depth Fisher	Neuron-trained PnP	0.219	0.195	-10.7% [-11.7, -9.7]
Worst-case coherence	TV	0.263	0.218	-17.0% [-17.9, -16.2]
Worst-case coherence	Pretrained PnP	0.378	0.290	-23.4% [-25.5, -21.1]
Worst-case coherence	Neuron-trained PnP	0.220	0.195	-11.0% [-12.2, -9.8]

Table S4. Objective comparison with the second optical sampling seed: 60 crops, $C = 8$, five draws per crop. W1 is in μm; relative changes compare our seed-0 plate with each seed-1 comparator, with paired cell-bootstrap 95 % intervals. Both seeds start from the same initial phase. Negative changes favor our criterion.

Comparator	Reconstruction	Other W1	Our W1	Change [95% CI]
Depth Fisher	TV	0.253	0.218	-13.9% [-15.7, -12.2]
Depth Fisher	Pretrained PnP	0.370	0.289	-22.0% [-24.8, -19.2]
Depth Fisher	Neuron-trained PnP	0.219	0.194	-11.5% [-12.9, -10.1]
Worst-case coherence	TV	0.263	0.218	-17.1% [-18.1, -16.0]
Worst-case coherence	Pretrained PnP	0.378	0.289	-23.7% [-25.9, -21.4]
Worst-case coherence	Neuron-trained PnP	0.220	0.194	-11.9% [-12.7, -10.8]

Table S5. Seed-matched sensitivity check for the second optical sampling seed. Each criterion is paired with our seed-1 plate on the same 60 crops and five draws. W1 is in μm; negative changes favor our criterion. Brackets are paired cell-bootstrap 95 % intervals.

plate at learning rate 10^{-3}. Plate and decoder are then updated jointly for 1200 plate steps. The plate learning rate starts at 0.05 and follows a cosine decay to 0.005 over the last 30 % of steps. Decoder learning rate is 3×10^{-4}; each joint step is followed by four decoder-only steps on fresh batches using the same direction subset and a detached operator. The phase outside the trained central region remains fixed.

The initial 1200-step run did not meet a prespecified plateau criterion: validation MSE changed by 3.2 % over its last 300 plate updates, above the 2 % threshold. Before test evaluation, a fixed continuation of 600 updates was specified, with phase learning rate 0.005, unchanged decoder recipe, fresh optimizer states and restored model parameters. The continuation's last-300-step change was 0.025 %. The selected phase checkpoint, at update 1500, minimizes validation MSE across the continuation checkpoints. Phase RMS change from initialization is 1.44 radians. These checks limit selection and verify training behavior; they are not a claim of global convergence.

Numerical checks. At initialization, differentiable assembly reproduces stored operator rows to relative error 8.5×10^{-8} after accounting for the subset/full-scan normalization scalar. An adjoint check and three central finite-difference gradient projections pass; directional derivatives agree to 0.03–0.07 %. A short descent check confirms that the phase changes and the loss decreases. Cross-server reconstruction checks compare recorded image-quality and centroid metrics rather than requiring bit-identical floating-point builds. These checks precede transfer comparisons.

Decoder comparison after optical training. The phase is fixed before evaluation with TV, pretrained PnP or neuron-trained PnP. Each operator uses its validation-selected scalar parameters, while denoisers remain unchanged. For the feed-forward comparison, every plate receives 3000 additional decoder-only steps at learning rate 3×10^{-4}, with validation every 500 steps. The untrained and model-only plates start this stage from the same warmed decoder; the end-to-end plate starts from its selected jointly trained checkpoint. The selected end-to-end checkpoint includes $1500 \times 5 = 7500$ joint-stage decoder updates. Including warm-up and final fine-tuning, it receives 15,500 updates, versus 8000 for every other plate. The untrained decoder receives all 8000 on its own operator; the other fixed-plate decoders receive 3000 on theirs. Architecture and final fine-tuning match, while total decoder training does not. The model with lowest validation MSE is used for each plate.

Feed-forward outputs are retained without final clipping for PSNR, foreground SSIM and the implemented centroid metric. Axial-profile W1 uses their nonnegative part, as specified in Sec. S5. This differs from the final clipping of TV/PnP outputs and is shared across all plates within the feed-forward comparison.

Comparison scope and timing. The primary end-to-end comparison uses one optical initialization and one supervised

training run. Its plate and decoder were optimized for MSE; the primary reported axial metric is W1. On the RTX 5080, decoder warm-up, joint training and continuation took about 4500, 11,150 and 5574 seconds, respectively; per-plate fine-tuning took about 2700 seconds. The approximately 20-minute model-only timing was measured on an RTX 4500 Ada. We do not infer a controlled runtime speedup from these different machines.

The criterion and end-to-end evaluations use the same neuronal test crops and matched acquisition conditions. Each comparison retains the signed differences under every reconstruction method. Full-scan and reduced-dose comparisons include all three iterative methods, as reported in Sec. S8.

S4. Reconstruction and validation

Shared framework. For TV and PnP, the forward operator and validation-selected scalar parameters change across encoders, while denoiser weights and solver families remain fixed. The separately fine-tuned feed-forward route is specified in Sec. S3. TV uses $\frac{1}{2}\|\mathbf{A}\mathbf{x}-\mathbf{y}\|_2^2+\lambda\|\nabla\mathbf{x}\|_{2,1}$ with a box constraint. PnP uses a quadratic data update with denoising and optional sparsity branches. A quadratic data term penalizes residuals but does not guarantee membership in the hard measurement-interval set used in Sec. S9. Generic denoisers need not minimize an explicit regularization energy.

ADMM and plug-and-play updates. Let C_j map the volume to branch j, with auxiliary variable $\mathbf{v}_j$ and scaled dual $\mathbf{d}_j$. Each iteration computes

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \Big\{ \tfrac{1}{2}\|\mathbf{A}\mathbf{x}-\mathbf{y}\|_2^2 + \sum_j \tfrac{\rho_j}{2}\|C_j\mathbf{x}-\mathbf{v}_j^k+\mathbf{d}_j^k\|_2^2 \Big\}, \tag{S15}$$

$$\mathbf{v}_j^{k+1} = \mathcal{F}_j(C_j\mathbf{x}^{k+1}+\mathbf{d}_j^k), \tag{S16}$$

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + C_j\mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}. \tag{S17}$$

The data update is solved by conjugate gradients. For TV, C_j is the 3D gradient or identity, and $\mathcal{F}_j$ is vector soft thresholding or box projection. PnP uses identity branch maps and denoisers in place of proximal functions. If P_o rearranges a shared volume into slices of orientation $o \in \{xy, yz, zx\}$, the slice branch is $P_o^{-1}DP_o$, with D acting independently on each slice. Pretrained PnP has these three branches; neuron-trained PnP replaces their shared 2D network and adds a 3D denoising branch. An optional ℓ_1 branch uses scalar soft thresholding. Generic DnCNN branches are not asserted to be proximal maps of a known joint regularizer.

TV. We use isotropic 3D forward-difference TV and ADMM with separate gradient and box branches, both with unit penalty. The box branch projects its auxiliary variable at each iteration, and the final volume is clipped. There are 300 ADMM iterations; each conjugate-gradient volume update uses at most 20 iterations at relative tolerance 10^{-5}. The starting volume is normalized back-projection clipped to $[0, 1]$. The regularization weight is selected from $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$ using mean validation PSNR. Edge selections extend the grid by a factor of three, at most twice. At $C = 8$, validation selects $\lambda = 0.1$ for all five primary designs. These fixed iteration budgets are not claims of exact minimization. Additional checks with 1000 iterations changed mean PSNR by -0.03 dB at $C = 8$ and centroid error by at most $0.001\,\mu$m; checks at $C = 1$ changed PSNR by at most 0.03 dB and centroid error by $0.006\,\mu$m.

PnP. Both variants run 20 ADMM iterations, with at most 20 conjugate-gradient steps per volume update and relative tolerance 10^{-5}. Denoising-branch penalties are $5s$ and the sparsity penalty is $10s$. The implementation sets nominal branch weights of 0.2 for the slice denoisers and 1.0 for the volumetric denoiser. The fixed denoisers ignore this proximal-scale argument, so these values activate the branches rather than adjusting denoising strength. Their coupling to the data is controlled by the shared penalty. The sparsity weight is $0.2l$; branches with zero weight are omitted. In the primary $C = 8$ comparison, pretrained PnP uses $s = 0.25, l = 0$ for every design. Neuron-trained PnP uses $s = 0.125$ for the untrained plate and $s = 0.25$ for the other designs; $l = 1/16$ only for the seed-0 Fisher plate and zero otherwise. We validate $s \in \{2^{-4}, 2^{-3}, \ldots, 4\}$ and $l \in \{0, 1/16\}$. Boundary selections extend the scale grid by a factor of two and the nonzero sparsity scale by four, at most twice. Final selections are interior. All PnP results start from normalized back-projection and are clipped to $[0, 1]$ only after the final iteration.

Denoisers. Pretrained PnP uses SCICO's 17M natural-image DnCNN (17 layers, trained at noise standard deviation 0.1) [2, 15], independently along three slice orientations. Its denoising noise level is fixed; it does not receive a specimen-dependent noise estimate. Neuron-trained PnP replaces this 2D model and adds a 3D branch. The 2D network has 17 layers and 64 channels; the 3D network has 12 layers and 48 channels. We do not evaluate 2D-only and 3D-only neuron-trained variants, so the comparison changes both the training distribution and denoising architecture.

Training uses 459 crops from cell 1, stacks 8 and 9 of animal SOM009, with stack 10 held out for monitoring. Duplicated planes and planes whose mean falls below half their neighbors' median are excluded before crop extraction. Both networks use Adam at 10^{-3}, Gaussian noise $\sigma \sim \mathcal{U}(0, 0.25)$, flips, lateral transpositions and intensity scaling in $[0.3, 1]$. The 2D model uses random slices for 30,000 steps at batch

size 64; the 3D model uses volumes for 15,000 steps at batch size 8. Neither contributes test data or is updated during optical training or reconstruction.

Parameter selection. The two vascular validation crops come from another animal and are used for all neuronal comparisons. Parameters are selected separately by encoder, compression and changed noise condition, with the same selection rule and fixed prior weights. No neuronal test crop is used for tuning. Model-only optical design uses neither validation nor test specimens. The supervised counterpart uses its separately held-out neuronal validation volumes as described in Sec. S3. Reference arrays are used only for metric reporting, not in the reconstruction updates.

S5. Data, metrics and paired statistics

Dataset partitions. The neuronal references are public two-photon stacks used in DEEP^2 [13]. Raw sampling is $0.25\,\mu\mathrm{m}$ laterally and $1\,\mu\mathrm{m}$ axially. A $32 \times 128 \times 128$ crop is laterally area-resampled to $32 \times 96 \times 96$ and normalized to unit peak. Its planes are assigned the $0.32\,\mu\mathrm{m}$ analysis spacing, giving $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$: lateral extent changes slightly and axial extent is compressed by a factor of 3.125. Reconstruction errors use these simulation coordinates. Test cells are SOM004 cells 1–2, SOM005 cell 1 and SOM006 cells 1–3: six cells from three animals. Each cell contributes two prespecified crops selected by projected foreground area times depth spread. After fixing reconstruction parameters, eight additional eligible crops per cell were sampled uniformly by location, excluding the two already selected. The resulting 48 crops were reconstructed without retuning. There are 60 test crops in the main comparison; other full-field studies use the 12 prespecified crops unless stated otherwise.

Denoisers and supervised optical/decoder models are trained on SOM009, a fourth animal, as specified in Secs. S3 and S4. TV/PnP parameter selection uses two vascular crops from a different animal; supervised checkpoint selection uses 48 held-out neuronal volumes from SOM009. Conformal calibration uses a held-out stack of SOM009 and does not tune the reconstruction. No test volume enters optical training, denoiser training, scalar tuning or interval calibration.

Profile and centroid errors. Let $g_c = \sum_z x_{\mathrm{gt}}(c, z)$ and retain columns with $g_c > 0.1 \max_{c'} g_{c'}$. We set $\pi_c = g_c / \sum_{c'} g_{c'}$ over retained columns and normalize each nonnegative profile to unit mass. The one-dimensional W1 distance equals Δz times the sum of absolute CDF differences; the main paper averages it with π_c. The implementation clips negative values to zero for the profile metric and divides by column mass plus 10^{-12}. A zero-mass estimate therefore remains a zero profile rather than being assigned a fabricated depth distribution. The centroid error uses the same reference columns and weights, but the feed-forward evaluation computes centroids from raw network outputs; the W1 metric uses their nonnegative part. TV/PnP estimates are clipped before both metrics. For positive, normalized profiles, W1 is at least their absolute centroid difference; this inequality is not asserted for signed feed-forward centroids. A centroid can match when the profile does not, but similar reductions in the two errors do not establish a larger gain in profile shape.

PSNR uses the complete volume and SCICO's default reference range, $\max(\mathbf{x}_{\mathrm{gt}}) - \min(\mathbf{x}_{\mathrm{gt}})$ [2]. Foreground SSIM uses a Gaussian-weighted 3D SSIM map ($\sigma = 1.5$, intensity range one), averaged over reference voxels above 10 % of peak intensity after two binary-dilation steps. These intensity metrics complement W1, which normalizes each column and therefore does not penalize brightness errors by itself.

Depth-blind references. The flat estimate preserves each column's total brightness but distributes it uniformly over depth. The depth-blind oracle preserves that brightness and uses the reference crop's global axial profile for every column. The latter has access to the correct projected structure and global depth distribution but no local depth placement. On the 12 prespecified crops, their W1 errors are 2.266 and $1.432\,\mu\mathrm{m}$, with centroid errors 2.148 and $1.322\,\mu\mathrm{m}$, respectively. They are context for interpreting axial metrics, not reconstruction methods supplied with the actual measurements.

Pairing and uncertainty. For each setting, encoders share the specimen, direction subset, nominal dose and noise-generation protocol. We average the five draws within each crop. A relative change is the ratio of learned and comparator crop means minus one, not the mean of per-crop percentages. Confidence intervals use 10,000 cell-bootstrap resamples, keeping all crops of a sampled cell together. Cells from the same animal may also be dependent; six cells from three animals do not support a precise population-level guarantee. The intervals describe this finite evaluation and are not corrected for all exploratory comparisons.

Consistent reconstruction metrics. W1, centroid, PSNR and foreground SSIM in Tabs. S2, S6 and S9 are computed from the same completed reconstruction rows. Each complete encoder/method configuration contains all 60 crops and five draws. The primary three-plate/four-method table contains 3,600 unique rows. The primary five-design/four-method table contains 6,000 unique rows, including the Fisher and coherence designs. Each criterion configuration has the same 60 crops and five draws. The second optical sampling seed adds 2,700 rows for our, Fisher and coherence

plates under TV/PnP, using the same crops. Ancillary 12-crop studies are labeled separately. The reconstruction reruns use the recorded operators, parameters, directions and noise. Relative to earlier runs, the largest per-row differences are $0.019\,\mathrm{dB}/0.0004\,\mu\mathrm{m}$ for TV and $0.198\,\mathrm{dB}/0.0204\,\mu\mathrm{m}$ for PnP (PSNR/centroid); mean metrics agree at the reported precision. The all-column W1 evaluation used a first-row reproduction check for each configuration before completing its remaining solves. We report the resulting numerical estimates rather than assuming bitwise reproducibility across GPU runs.

S6. Full reconstruction results

Three optical designs and four reconstruction methods. Table S6 reports W1, centroid error, PSNR and foreground SSIM from the same 3,600 reconstructions: three plates, four methods, 60 crops and five draws. The feed-forward decoder is fine-tuned separately for each plate, whereas TV and the PnP networks are independent of optical training. Table S7 gives paired cell-bootstrap intervals for W1. Negative changes favor the first named design. The end-to-end plate has a measurable advantage with its own decoder and under both PnP methods, while our plate has lower W1 under TV. The different directions preclude a universal ranking of the two learned plates.

Optical sampling repeats. Table S9 adds our second optical sampling seed to the primary TV/PnP comparison, using the same untrained initialization. Each our plate lowers W1 on every crop under each method. Seed 0 improves centroid error on 59 of 60 crops for neuron-trained PnP and all crops for the other methods. Seed 1 improves centroid error on all crops. These repeats test sampling variability, not independence of phase initialization.

Reference-based and random crops. Table S10 separates the 12 prespecified crops from the 48 later random crops. All methods retain positive optical gains on the latter, with parameters already fixed. This addresses dependence on the initial structure-score selection, while preserving the cell and animal limitations of the dataset.

Comparison with untrained encoders. The separate centroid study in Tab. S11 uses pretrained PnP, the 12 prespecified crops and the expanded validation grid. Its phase, amplitude and DMD rows share the same $2\times$ aperture, relay, photon model, direction draws and tuning procedure. Learned centroid errors of 0.230–$0.234\,\mu\mathrm{m}$ are below all three random phase plates (0.422–$0.438\,\mu\mathrm{m}$) and amplitude masks (0.396–$0.523\,\mu\mathrm{m}$). The learned seed-0 plate reduces centroid error by 42–56 % against the amplitude masks and 45–47 % against the random phase plates. This shows that its initialization is representative of these controls. It does not show that learning succeeds from other initializations.

The two in-focus DMD pattern sets remain close in centroid error and image quality. On the same 12 prespecified crops, the primary DMD control gives W1 of 2.375, 2.238 and $2.270\,\mu\mathrm{m}$ under TV, pretrained PnP and neuron-trained PnP, respectively, close to the $2.266\,\mu\mathrm{m}$ uniform-depth reference. Their low depth discrimination concerns this fixed-focus control, not all amplitude encoders. Indeed, the defocused amplitude masks recover depth and can match random phase plates in centroid error. The plate-free scan also retains weak finite-aperture diversity, so its observations are not an ideal zero-depth-information system.

Point-emitter diagnostics. For 25 interior lateral locations and 16 depths, we compare measurement columns under a Gaussian approximation of shot noise with unknown emitter brightness. Random phase plates assign the wrong depth in 1.5–1.7 % of cases with 32 directions; amplitude masks give about 5 %, the in-focus DMD about 25 %, and the plate-free control about 89 %. These are model-based emitter diagnostics, not measured resolution of dense structures. The plate-free diagnostic uses the original $2\times$ aperture; its neuronal control in Tab. S11 uses $4\times$.

Comparisons across priors. A descriptive pairing of our plate + TV against untrained plate + neuron-trained PnP gives 19.6 % lower W1 (95 % CI: 16.4–22.5 %), improving 56 of 60 crops. It does not establish a general dominance relation: our plate + pretrained PnP has 7 % higher mean W1 than untrained plate + neuron-trained PnP (interval -2.6 to $+17.3$ %). Optical and prior choices interact; the comparisons in this paragraph concern only the untrained plate and our design, not the end-to-end plate.

S7. Structural tasks and measurement discrimination

Crossing phantoms. We build Gaussian fluorescent tubes on the $32 \times 96 \times 96$ analysis grid, at $0.32\,\mu\mathrm{m}$ spacing. Their lateral and axial standard deviations are $0.32\,\mu\mathrm{m}$, with half-length $12.8\,\mu\mathrm{m}$. The equal-branch reconstruction phantoms combine tubes by their maximum before $2 \times 2 \times 2$ averaging. Three lateral geometries use filament-a orientation/crossing-angle pairs $(0°, 90°)$, $(30°, 60°)$ and $(60°, 45°)$. Centers are reconstruction planes 7 and 5. Equal-branch gaps are 0, 2, 3, 4, 6 and 8 planes (0, 1.28, 1.92, 2.56, 3.84 and $5.12\,\mu\mathrm{m}$), giving 36 phantoms, each with five measurement draws. All use $C = 8$, matched nominal dose and parameters selected on the vascular validation crops, not on these phantoms.

Resolved peaks and arm ordering. Peaks are detected on zero-padded axial columns, requiring height and promi-

Reconstruction	Plate	W1 (μm)	Centroid (μm)	PSNR	FG-SSIM
TV	Untrained	0.371	0.259	36.70	0.850
TV	Ours	0.218	0.150	38.29	0.910
TV	End-to-end	0.222	0.151	38.07	0.905
Pretrained PnP	Untrained	0.579	0.490	35.72	0.806
Pretrained PnP	Ours	0.290	0.224	37.56	0.900
Pretrained PnP	End-to-end	0.281	0.212	37.52	0.897
Neuron-trained PnP	Untrained	0.271	0.197	38.01	0.902
Neuron-trained PnP	Ours	0.195	0.139	39.60	0.934
Neuron-trained PnP	End-to-end	0.189	0.131	39.70	0.936
Feed-forward	Untrained	0.366	0.282	36.67	0.872
Feed-forward	Ours	0.265	0.196	38.04	0.912
Feed-forward	End-to-end	0.234	0.167	38.53	0.922

Table S6. Three-plate comparison at $C = 8$ and matched nominal dose, with 60 crops and five draws. Errors are in μm. All metrics in a row use the same reconstructions. Feed-forward networks are fine-tuned for their respective plates; their earlier training histories are specified in Sec. S3.

Reconstruction	Ours vs. untrained	End-to-end vs. ours
TV	-41.3% [-44.5, -37.7]	$+1.9\%$ [$+1.1$, $+2.7$]
Pretrained PnP	-49.9% [-52.8, -46.6]	-3.0% [-5.0, -1.0]
Neuron-trained PnP	-27.9% [-29.5, -26.0]	-3.2% [-5.6, -1.2]
Feed-forward	-27.5% [-31.2, -24.7]	-11.6% [-14.4, -9.0]

Table S7. Paired relative W1 changes (95 % cell-bootstrap intervals) for the same 60-crop comparison. Each change is the ratio of crop-mean errors minus one, after averaging the five draws.

Reconstruction	W1 difference [95% CI]	Centroid difference [95% CI]
TV	$+0.004$ [$+0.002$, $+0.006$]	$+0.001$ [-0.001, $+0.004$]
Pretrained PnP	-0.009 [-0.014, -0.003]	-0.012 [-0.018, -0.006]
Neuron-trained PnP	-0.006 [-0.011, -0.002]	-0.007 [-0.012, -0.004]
Feed-forward	-0.031 [-0.037, -0.025]	-0.029 [-0.036, -0.023]

Table S8. End-to-end minus our plate: absolute error differences and paired cell-bootstrap 95 % intervals (μm), on the primary 60 crops. Mean W1 differences are below 0.01 μm under the iterative methods; this is not a bound on their intervals or individual-crop differences.

nence of at least 20 % of that column's maximum. Reference and reconstructed peaks are matched one-to-one within one reconstruction plane, closest first. Crossing columns must carry more than 20 % of each reference filament's maximum column signal and contain reference peaks matched to both planes. The resolved fraction is the share of reference crossing-column signal in columns whose reconstruction has both matched peaks. It is not an absolute recovered-brightness cutoff.

For ordering, each filament's arm columns carry at least 20 % of its own maximum column signal and less than 2 % of the other filament's. Their reconstructed signal-weighted axial centroids must follow the true order, with consecutive arms separated by more than half a plane. A volume succeeds when the resolved fraction is at least 0.5 and ordering is correct. In the zero-gap control, two matched peaks instead count as a false split.

The qualifying gap is the smallest tested gap of at least two planes with success rate at least 0.8 at that gap and every larger tested gap, at both centers separately. Each center's rate uses 15 reconstructions (three geometries and five draws). The main figure pools the two centers in each curve; the criterion uses their individual rates.

The displayed example uses the first lateral geometry, center 7 and draw 0. Its gap was chosen at our plate's TV threshold, so the gap is outcome-informed. Untrained and learned TV succeed in 10 % and 90 % of the pooled cases at that gap; the displayed draw is one failed/successful pair. All displayed sections share geometry and intensity scale. Neither plate falsely splits the merged control.

Reconstruction	W1 (μm)	Centroid (μm)	PSNR	FG-SSIM	W1 change [95% CI]
TV	0.218	0.149	38.31	0.910	-41.3% [-44.6, -37.5]
Pretrained PnP	0.289	0.222	37.58	0.900	-50.1% [-53.1, -46.7]
Neuron-trained PnP	0.194	0.136	39.64	0.934	-28.5% [-30.1, -27.1]

Table S9. Our second optical sampling seed at $C = 8$, 60 crops and five draws per crop, with matched nominal dose. Changes compare it with the shared untrained initialization; brackets are cell-bootstrap 95 % intervals. Seed 0 is in Tab. S6.

Reconstruction	Seed	Crops	W1 change [95 % CI]	W1 improved	Centroid change [95 % CI]
TV	0	12 prespecified	-38 % [-43, -34]	12/12	-37 % [-41, -34]
TV	0	48 random	-42 % [-46, -37]	48/48	-43 % [-49, -38]
TV	1	12 prespecified	-39 % [-42, -36]	12/12	-38 % [-40, -36]
TV	1	48 random	-42 % [-46, -37]	48/48	-44 % [-49, -38]
Pretrained PnP	0	12 prespecified	-44 % [-50, -39]	12/12	-46 % [-53, -41]
Pretrained PnP	0	48 random	-51 % [-54, -47]	48/48	-56 % [-59, -52]
Pretrained PnP	1	12 prespecified	-44 % [-49, -38]	12/12	-46 % [-52, -40]
Pretrained PnP	1	48 random	-51 % [-55, -47]	48/48	-57 % [-60, -52]
Neuron-trained PnP	0	12 prespecified	-27 % [-34, -20]	12/12	-26 % [-35, -18]
Neuron-trained PnP	0	48 random	-28 % [-30, -26]	48/48	-31 % [-34, -28]
Neuron-trained PnP	1	12 prespecified	-29 % [-34, -25]	12/12	-29 % [-35, -24]
Neuron-trained PnP	1	48 random	-28 % [-30, -27]	48/48	-31 % [-34, -29]

Table S10. Separate evaluation on the prespecified and randomly sampled crops. Negative changes favor our plate. Both sets contain crops from the same six cells; the 48 additional crops expand spatial coverage rather than biological replication.

Encoder	Centroid (μm)	PSNR (dB)	FG-SSIM
Ours, seed 0	0.230	36.15	0.868
Ours, seed 1	0.234	36.21	0.869
Untrained phase 0	0.431	34.81	0.784
Untrained phase 1	0.422	34.93	0.787
Untrained phase 2	0.438	34.85	0.782
Amplitude mask 0	0.523	33.70	0.703
Amplitude mask 1	0.419	34.37	0.766
Amplitude mask 2	0.396	34.36	0.763
DMD set 0	2.169	28.83	0.482
DMD set 1	2.169	28.81	0.479
Plate-free, $4\times$ aperture	1.102	30.18	0.494

Table S11. Untrained-encoder control study at $C = 8$: 12 prespecified crops, pretrained PnP. W1 was not computed in this study. The plate-free row uses a $4\times$ aperture in a separate study; its own random-plate control gives $0.446\,\mu$m/34.82 dB. Other rows use $2\times$ apertures. Do not interpret the plate-free row as an exactly matched cross-aperture contrast.

Weak and three-filament extensions. The weak-branch cases use the same two-filament geometries with the deeper branch at 0.3 times the first's intensity. Gaps are 0, 3, 4, 6 and 8 planes; at two planes the reference itself fails the peak criterion. Only TV with our plate meets the two-center threshold, at $3.84\,\mu$m (Tab. S12).

Three equal filaments occupy planes $c - d, c, c + d$, with orientations $(0°, 60°, 120°)$ or $(30°, 90°, 150°)$. Initially tested spacings $d = 0, 2, 3, 4$ planes do not achieve 80 % success. A descriptive extension adds five planes at both centers and six at center 7, using the same parameters and success rule. Our plate then resolves all three filaments at five-plane spacing ($3.20\,\mu$m) under TV and pretrained PnP at both centers. The untrained plate reaches 80 % only at center 7: six planes under TV and five under pretrained PnP. With neuron-trained PnP, our plate reaches the criterion at six planes at center 7 but not at center 5; the untrained plate reaches it at neither center. These extensions show the dependence on spacing and prior, rather than a general guarantee for denser structures.

	Equal branches		Weak branch		Three filaments	
Reconstruction	Untrained	Ours	Untrained	Ours	Untrained	Ours
TV	3.84	2.56	–	3.84	–	3.20
Pretrained PnP	3.84	2.56	–	–	–	3.20
Neuron-trained PnP	3.84	3.84	–	–	–	–

Table S12. Qualifying gaps (μm) under the same success rule. A dash means the criterion is not reached at both centers over the tested gaps. Weak branch: the second filament has 0.3 times the first's intensity. The three-filament columns include the descriptive wider-gap extension: $3.20\,\mu$m was tested at both centers, and $3.84\,\mu$m only at center 7, since the other geometry would leave the field.

Joint-training control. The end-to-end plate is evaluated on the same equal-branch phantoms under all three iterative methods. Pooling the five positive gaps, its success rate minus ours is -3.3 percentage points under TV (95 % paired phantom-bootstrap interval: -8.7 to $+1.3$), $+0.7$ under pretrained PnP (-1.3 to $+2.7$), and -0.7 under neuron-trained PnP (-5.3 to $+4.7$). Full-dose differences are not established (Tab. S13); Fig. S3 shows all gap curves. The zero-gap controls show no false splitting under any of these plate–method combinations. Reduced-dose comparisons are given in Sec. S8.

Merged neuronal alternatives. A reference column is called layered when it belongs to the signal mask and has at least two peaks under the same relative height/prominence rule. For each corresponding fine-grid lateral column, we move its intensity to its axial centroid, distributing mass between the two neighboring planes to preserve that centroid exactly. Other columns remain unchanged. The resulting alternative preserves the fine-grid lateral projection and each modified column's centroid while changing its depth profile. One crop has no eligible columns, leaving 59. These are constructed candidates, not solutions optimized over a measurement-consistency set; concentrated intensities need not remain below the original normalized peak.

For noise-free expected counts μ of the reference and $\nu(c)$ of a brightness-scaled alternative, we compute

$$D_{\mathrm{P}}(\mu\|\nu) = \sum_i \left[\mu_i \log \frac{\mu_i}{\nu_i} + \nu_i - \mu_i\right]. \qquad \text{(S18)}$$

The simulator evaluates both objects on the fine grid, using each matched direction subset and the same exposure. Without background, the profiled scale is the ratio of total reference to alternative expected counts. With background, the numerical fit searches a log-scale interval of $[-1, 1]$ around the total-signal estimate. Crop scores average five subsets. The geometric mean of the crop-level learned/untrained ratios is 9.99 (95 % CI: 8.66–11.79) without background and 10.4 with background. Against the end-to-end plate, our ratio is 2.00 (1.91–2.09), with higher divergence on all 59 eligible crops. The draw-average is taken within each crop before forming the ratios. Untrained-plate divergences can already be large; the ratio is a diagnostic of these specified alternatives, not a probability of successful reconstruction. No KL-to-reconstruction-success guarantee is claimed.

Additional neuronal structure metrics. Layered columns carry about 17 % of reference signal on average. Table S14 uses the completed primary reconstruction rows and the 59 crops with eligible columns; one crop has none. Relative to the untrained plate, our plate lowers their profile W1 by 34 % with TV, 42 % with pretrained PnP and 28 % with neuron-trained PnP. The end-to-end plate has higher W1 than ours under TV, no established difference under either PnP method, and lower W1 with its own decoder. These descriptive subset results contextualize the single-column figures. Most eligible columns contain intensity structure within thick branches, so lower profile error does not establish separation of distinct neuronal branches or a larger benefit than in centroid accuracy.

A prespecified TV comparison did not establish a larger relative profile gain than centroid gain on layered columns: the difference was -0.7 percentage points (95 % CI: -2.9 to $+1.5$). PnP subset statistics remain descriptive because an earlier prespecified PSNR reproduction tolerance of 0.05 dB was not met; the completed full-field reruns use the numerical checks documented in Sec. S5.

On all 60 crops, we also report centerline Dice [12], which compares a reference mask at 10 % of peak intensity with an estimated mask retaining the same number of voxels, followed by 3D skeletonization. Scores rise from 0.79 to 0.84 for TV, 0.75 to 0.84 for pretrained PnP and 0.85 to 0.88 for neuron-trained PnP. This thresholded metric is depth-sensitive but does not substitute for branch tracing or connectivity validation.

S8. Measurement, photon and alignment controls

Dose convention. Each encoder's excitation is divided by its mean over the specimen analysis grid and the full 256-direction scan. This fixes the deposited excitation per complete scan. For k directions, exposure is multiplied by

Reconstruction	Ours (%)	End-to-end (%)	Difference (points)
TV	62.0	58.7	-3.3 [-8.7, $+1.3$]
Pretrained PnP	64.0	64.7	$+0.7$ [-1.3, $+2.7$]
Neuron-trained PnP	44.0	43.3	-0.7 [-5.3, $+4.7$]

Table S13. Full-dose crossing success pooled across 30 positive-gap cases, with five draws per case. Differences are end-to-end minus ours, with paired phantom-case bootstrap 95 % intervals.

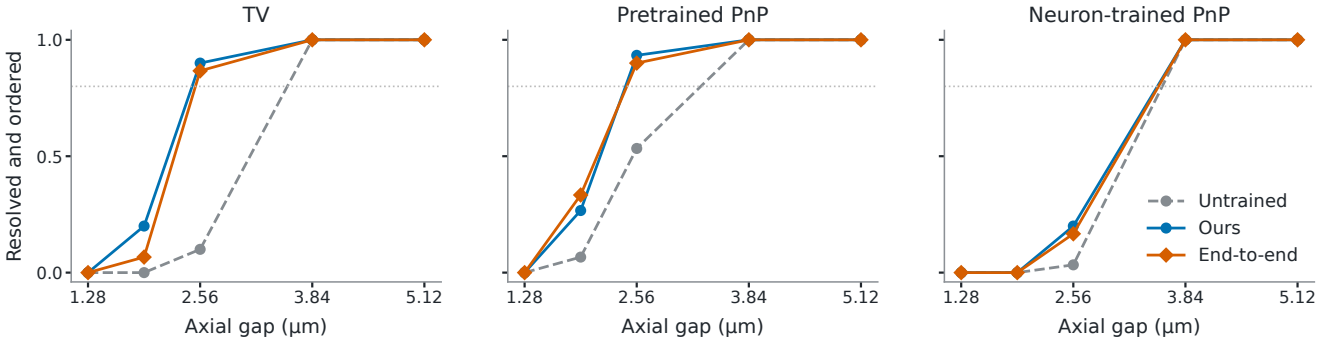


Figure S3. Full-dose equal-branch crossing curves for all three iterative methods, at $C = 8$ and matched nominal dose. Each point pools three lateral geometries, two axial centers and five draws. The qualifying-gap rule requires at least 80 % success at each center separately and at all larger tested gaps. Under neuron-trained PnP, both the untrained and our plates qualify at 3.84 μm. Zero-gap controls have no false splitting.

Reconstruction	Untrained	Ours	End-to-end	Ours vs. untrained	End-to-end minus ours
TV	0.467	0.309	0.318	-33.8% [-36.3, -31.7]	$+0.009$ [$+0.004$, $+0.015$]
Pretrained PnP	0.641	0.369	0.370	-42.4% [-46.5, -38.5]	$+0.001$ [-0.008, $+0.009$]
Neuron-trained PnP	0.374	0.271	0.272	-27.6% [-28.8, -26.4]	$+0.001$ [-0.002, $+0.004$]
Feed-forward	0.518	0.387	0.352	-25.4% [-29.6, -21.8]	-0.035 [-0.045, -0.028]

Table S14. Profile W1 on layered reference columns: 59 eligible crops, $C = 8$, five draws per crop, using the same completed reconstructions as the primary table. Errors and absolute end-to-end-minus-ours differences are in μm; brackets are paired cell-bootstrap 95 % intervals. Feed-forward decoders have the training histories in Sec. S3.

$256/k$. Random subsets therefore match nominal dose in expectation, not exactly for each subset. The per-specimen exposure is set from the untrained plate's full-scan mean signal and reused for every encoder. Equal deposited dose does not force identical detected counts: illumination can correlate differently with the specimen. For our two optical sampling seeds, mean detected signal changes by about -1 to -2 % across the 12 crops, with individual changes approximately -12 to $+9$ %.

Measurement compression. Table S15 reports both depth metrics at $C = 1, 4, 8$. These values count all camera samples, not camera images alone. At matched nominal dose, fewer directions mean longer exposure per image. The learned 32-image acquisition gives lower W1 than the 256-image random acquisition for TV and pretrained PnP, but the close neuron-trained-PnP means do not establish equivalence (Tab. S16). The TV curve is not monotone in compression: validation can prefer a stronger regularization weight at $C = 4$, which smooths neuronal features more than at $C = 8$.

Full-scan joint-training comparison. With all 256 directions ($C = 1$), our plate has lower neuronal W1 than the end-to-end plate under pretrained PnP; differences under TV and neuron-trained PnP are not established (Tab. S17). The 12 crops and exposure protocol are shared, with one complete scan per crop. This complements the primary compressed-scan comparison.

Noise conditions. The nominal model uses Poisson shot noise. The background condition adds a uniform mean equal to the untrained plate's mean expected signal for the same exposure, plus independent Gaussian read noise with standard deviation 2 electrons. The known mean background is subtracted before reconstruction. The same background

Reconstruction	Plate	W1, $C = 1$	Centroid	W1, $C = 4$	Centroid	W1, $C = 8$	Centroid
TV	Untrained	0.249	0.187	0.298	0.223	0.318	0.243
TV	Ours, seed 0	0.170	0.135	0.208	0.166	0.196	0.153
TV	Ours, seed 1	0.160	0.124	0.202	0.158	0.195	0.151
Pretrained PnP	Untrained	0.413	0.357	0.445	0.384	0.496	0.431
Pretrained PnP	Ours, seed 0	0.222	0.184	0.254	0.212	0.277	0.231
Pretrained PnP	Ours, seed 1	0.213	0.173	0.249	0.206	0.279	0.234
Neuron-trained PnP	Untrained	0.203	0.161	0.234	0.184	0.282	0.221
Neuron-trained PnP	Ours, seed 0	0.142	0.110	0.167	0.130	0.206	0.163

Table S15. Measurement budgets on the 12 prespecified crops. W1 and centroid errors are in μm and come from the same reconstruction rows. $C = 1$ uses all 256 directions; $C = 4, 8$ average five subsets of 64 and 32 directions. The second seed is available across these budgets for TV and pretrained PnP; neuron-trained PnP is evaluated with seed 0 here and both seeds at $C = 8$.

Acquisition comparison	Reconstruction	Ours	Untrained	Change [95% CI]
32 vs. 256 images	TV	0.196	0.249	-21.4% $[-27.9, -14.5]$
32 vs. 256 images	Pretrained PnP	0.277	0.413	-32.9% $[-40.4, -25.7]$
32 vs. 256 images	Neuron-trained PnP	0.206	0.203	$+1.7\%$ $[-12.1, +13.0]$
0.3 vs. nominal dose	TV	0.234	0.318	-26.4% $[-32.7, -19.8]$
0.3 vs. nominal dose	Pretrained PnP	0.302	0.496	-39.2% $[-45.1, -33.7]$
0.3 vs. nominal dose	Neuron-trained PnP	0.220	0.282	-22.0% $[-28.3, -16.4]$

Table S16. Acquisition trade-offs on the 12 prespecified crops. Values are W1 in μm; changes compare our plate with the untrained acquisition specified in each row, with paired cell-bootstrap 95 % intervals. The image-count comparison uses matched nominal dose; the dose comparison uses $C = 8$ for both plates.

Reconstruction	Ours	End-to-end	Difference [95% CI]
TV	0.170	0.177	$+0.007$ $[-0.001, +0.014]$
Pretrained PnP	0.222	0.231	$+0.009$ $[+0.003, +0.016]$
Neuron-trained PnP	0.142	0.143	$+0.001$ $[-0.005, +0.005]$

Table S17. Full-scan comparison on the 12 prespecified crops at $C = 1$ and matched nominal dose. W1 and end-to-end-minus-ours differences are in μm, with paired cell-bootstrap 95 % intervals.

level, exposure and noise protocol are used for each encoder. A separate low-dose condition reduces exposure to one tenth. Scalar reconstruction parameters are revalidated for each changed noise condition, while networks stay fixed. Table S18 reports all three methods.

Reduced-dose joint-training comparison. At one tenth of the nominal dose, both learned plates are evaluated under all three iterative methods on the 12 prespecified neuronal crops and on equal-branch crossings. Networks remain fixed and scalar parameters follow the condition-specific validation protocol. On neuronal crops, the end-to-end plate has lower W1 under neuron-trained PnP (0.227 versus 0.239 μm), while differences under TV and pretrained PnP are not established (Tab. S19a). For crossings, our plate succeeds more often under TV and pretrained PnP when pooling the three positive gaps 1.28, 1.92 and 2.56 μm. Both plates have low success under neuron-trained PnP, without an established difference (Tab. S19b and Fig. S4). Each gap has six phantom cases (three geometries, two centers), with three draws per case. The paired bootstrap averages draws and resamples the 18 cases, rather than treating draws as independent phantoms. This comparison is a targeted structure stress test, not evidence of generally better low-dose neuronal reconstruction.

Photon budget. The dose sweep uses factors 0.1, 0.3, 1 and 3 of the nominal exposure (Tab. S20). At dose 0.3, our plate's W1 is 0.234, 0.302 and 0.220 μm for TV, pretrained PnP and neuron-trained PnP, compared with 0.318, 0.496 and 0.282 μm for the untrained plate at dose 1. These reductions of 26 %, 39 % and 22 % describe the tested simulation and inference budgets. The untrained curve's flattening between doses 1 and 3 is not evidence of a fundamental photon limit.

Condition	Reconstruction	W1, random	W1, learned	Centroid, random	Centroid, learned	W1 change	Centroid change
Shot noise	TV	0.318	0.196	0.243	0.153	−38 % [−43, −34]	−37 % [−41, −34]
Shot noise	Pretrained PnP	0.496	0.277	0.431	0.231	−44 % [−50, −39]	−46 % [−53, −41]
Shot noise	Neuron-trained PnP	0.282	0.206	0.221	0.163	−27 % [−34, −20]	−26 % [−35, −18]
Background + read noise	TV	0.324	0.227	0.249	0.180	−30 % [−38, −22]	−28 % [−36, −20]
Background + read noise	Pretrained PnP	0.627	0.297	0.548	0.247	−53 % [−57, −48]	−55 % [−60, −50]
Background + read noise	Neuron-trained PnP	0.288	0.210	0.228	0.166	−27 % [−35, −19]	−27 % [−36, −18]
0.1 dose	TV	0.466	0.292	0.365	0.236	−37 % [−45, −29]	−35 % [−44, −28]
0.1 dose	Pretrained PnP	0.834	0.407	0.733	0.352	−51 % [−57, −45]	−52 % [−58, −45]
0.1 dose	Neuron-trained PnP	0.345	0.239	0.269	0.186	−31 % [−37, −24]	−31 % [−38, −24]

Table S18. Noise controls on 12 crops at $C = 8$. Errors are in μm; changes and cell-bootstrap 95 % intervals compare our and untrained plates within the same condition.

(a) Neuronal axial-profile W1 (μm), 12 crops

Reconstruction	Ours	End-to-end	Difference [95% CI]
TV	0.292	0.285	−0.007 [−0.028, +0.011]
Pretrained PnP	0.407	0.405	−0.001 [−0.019, +0.015]
Neuron-trained PnP	0.239	0.227	−0.012 [−0.027, −0.001]

(b) Crossing success, 18 phantom cases

Reconstruction	Ours (%)	End-to-end (%)	Difference (points)
TV	16.7	0.0	−16.7 [−29.6, −5.6]
Pretrained PnP	38.9	24.1	−14.8 [−29.6, −1.9]
Neuron-trained PnP	5.6	0.0	−5.6 [−11.1, +0.0]

Table S19. Joint-training control at $0.1\times$ nominal dose and $C = 8$. Differences are end-to-end minus ours. Intervals resample cells for neuronal W1 and phantom cases for crossing success, after averaging measurement draws. Negative success differences favor our plate; negative W1 differences favor end-to-end.

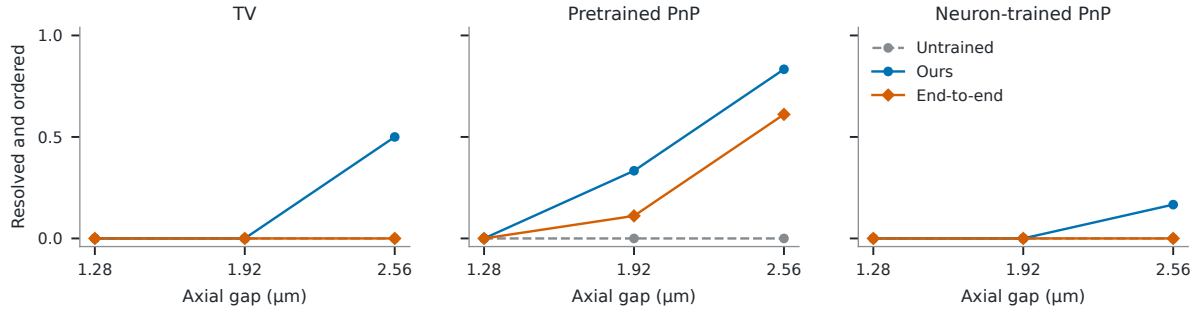


Figure S4. Full low-dose crossing curves for the tested positive gaps. Each point pools the two axial centers, three lateral geometries and three draws. At zero gap, none of the tested plate–method combinations falsely splits the merged structure.

Alignment. Distance perturbations set $D = 6.1$ or $7.1\,\mu$m. In the characterized case, the operator is rebuilt from the known distance and scalar reconstruction parameters are revalidated; the phase is unchanged. In the nominal-operator case, reconstruction ignores the perturbation. Lateral error shifts the plate by one $0.32\,\mu$m feature. Pointing errors are offsets of 0.005 or 0.01 in transverse direction cosine. All comparisons use neuron-trained PnP, $C = 8$, the same 12 crops and five draws.

Table S21 separates these assumptions. Known distance changes retain the optical gain. Ignoring distance or lateral registration can make our plate worse than the random one, including an 11 % W1 increase for a one-feature lateral shift. Rebuilding an operator with a known perturbation is not an experimentally demonstrated calibration procedure. Phase-quantization experiments are not used to claim a fabricable low-level plate: the simulated surround was not quantized with the trained center.

Reconstruction	Nominal dose	W1, random	W1, learned	Centroid, random	Centroid, learned
TV	0.1	0.466	0.292	0.365	0.236
TV	0.3	0.442	0.234	0.341	0.185
TV	1	0.318	0.196	0.243	0.153
TV	3	0.315	0.192	0.242	0.151
Pretrained PnP	0.1	0.834	0.407	0.733	0.352
Pretrained PnP	0.3	0.631	0.302	0.550	0.251
Pretrained PnP	1	0.496	0.277	0.431	0.230
Pretrained PnP	3	0.494	0.276	0.430	0.230
Neuron-trained PnP	0.1	0.345	0.239	0.269	0.186
Neuron-trained PnP	0.3	0.304	0.220	0.236	0.174
Neuron-trained PnP	1	0.282	0.207	0.221	0.164
Neuron-trained PnP	3	0.275	0.203	0.218	0.162

Table S20. Dose sweep at $C = 8$, 12 crops. W1 and centroid errors (μm) are computed from the same reconstructions. Each exposure factor is relative to the matched nominal-dose protocol.

Condition	W1, random	W1, learned	Centroid, random	Centroid, learned	W1 change	Centroid change
none (design)	0.282	0.206	0.221	0.163	$-27\,\%$ [-34, -20]	$-26\,\%$ [-35, -18]
Known: D=6.1	0.285	0.205	0.225	0.162	$-28\,\%$ [-35, -22]	$-28\,\%$ [-37, -20]
Known: D=7.1	0.284	0.209	0.226	0.166	$-27\,\%$ [-34, -20]	$-26\,\%$ [-35, -19]
Nominal operator: scan tilt 0.005	0.318	0.254	0.253	0.196	$-20\,\%$ [-25, -16]	$-22\,\%$ [-30, -15]
Nominal operator: scan tilt 0.01	0.379	0.328	0.308	0.251	$-13\,\%$ [-16, -10]	$-18\,\%$ [-22, -15]
Nominal operator: D=6.1	0.563	0.586	0.528	0.571	$+4\,\%$ [$+0.3$, $+8$]	$+8\,\%$ [$+3$, $+14$]
Nominal operator: D=7.1	0.447	0.473	0.383	0.438	$+6\,\%$ [$+5$, $+7$]	$+14\,\%$ [$+12$, $+17$]
Nominal operator: lateral shift 1 feature	0.451	0.500	0.372	0.400	$+11\,\%$ [$+2$, $+23$]	$+7\,\%$ [-0.4, $+19$]

Table S21. Alignment controls with neuron-trained PnP. Errors are in μm; changes compare our and untrained plates under the same perturbation. "Known" means the offset is supplied to operator assembly and validation; "nominal operator" means it is ignored. Positive changes favor the untrained plate.

S9. Complementary uncertainty analyses

These analyses ask different questions from the main reconstruction tests. Centroid intervals quantify an empirical depth error rate after reconstruction. Set-membership bounds assess admissible voxel intensities without a learned prior. Neither supplies a connectivity guarantee for the neuronal reconstructions.

Conformal centroid intervals. We form $\hat{z}_c \pm \lambda$ around each reconstructed axial centroid. For a volume, $L(\lambda)$ is the reference-signal-weighted fraction of evaluated columns whose true centroid lies outside the interval. It is bounded by one and decreases with λ. Conformal risk control [1] selects

$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \le \alpha\right\}, \qquad \text{(S19)}$$

where $\hat{R}_n$ averages the loss over n calibration volumes. Under exchangeability of the calibration volumes and a new volume, $\mathbb{E}[L(\hat{\lambda})] \le \alpha$. This controls expected signal-weighted miscoverage, not each column's or specimen's coverage.

We use neuron-trained PnP at $C = 8$, target $\alpha = 0.1$ and encoder-specific calibration on 35 non-overlapping crops from a held-out SOM009 stack. A 240-crop overlapping set is a sensitivity analysis, not independent replication. On the 12 prespecified test crops, our seed-0 half-width decreases from 0.646 to $0.474\,\mu$m (26.7 %), with empirical miscoverage 0.075 versus 0.060 for the untrained plate (Tab. S22). The test animals differ from the calibration animal, so exchangeability is not established; observed miscoverage is empirical rather than a cross-animal guarantee.

	35 non-overlapping crops		all 240 crops	
plate	$\hat{\lambda}$ (μm)	test miscov.	$\hat{\lambda}$ (μm)	test miscov.
random	0.646	0.060	0.570	0.085
learned, seed 0	0.474	0.075	0.422	0.091
learned, seed 1	0.486	0.060	0.422	0.081

Table S22. Calibrated centroid half-widths and empirical signal-weighted test miscoverage. The 240-crop analysis reuses overlapping regions. Seed 1 gives $0.486\,\mu$m on the primary 35-crop calibration set.

Voxel bounds without a learned prior. Given $\mathbf{x} \in [0,1]^N$, define

$$\mathcal{C}(\mathbf{y}) = \{\mathbf{x} : \ \ell_i \le (\mathsf{A}\mathbf{x})_i \le h_i,\ 0 \le \mathbf{x} \le 1\}. \qquad \text{(S20)}$$

For a nonempty set, minimizing and maximizing x_j gives the smallest and largest voxel intensities compatible with the stated constraints [6, 9]. We solve these linear programs with HiGHS [7]. The reconstruction's quadratic data term does not automatically place its output in this set.

The reduced field contains $4 \times 12 \times 12$ reconstruction voxels and 3×3 camera samples per image, using 32 fixed directions. Thus, $N = 576$ and $M = 288$; this is a separate geometry, not the full-field $C = 8$ experiment. Plates are trained separately on this reduced geometry; no reflected-depth pair qualifies for the full-grid weighting rule, so all off-diagonal weights are one.

The ancillary Gaussian model uses reference full-scan mean m_p for the untrained plate and mean m_e for an encoder, with $\sigma_e = m_p 10^{-q/10}\sqrt{k/256}\sqrt{m_e/m_p}$ and reference parameter $q = 15$. At $q = 15$, this is a spatially uniform Gaussian approximation to the main photon model at each encoder's mean signal, rather than a pixel-wise Poisson variance. The formula defines q explicitly; it is not a standard RMS SNR. Intervals use $y_i \pm z\sigma_e$, where $z = \Phi^{-1}(1 - 0.05/(2 \cdot 2880))$. The family contains all 2304 rows of the 256-direction reduced scan plus up to 576 projection constraints. Using the same correction for every subset gives at least 95 % simultaneous coverage under the stated Gaussian model and exact reduced linear operator. Unmodeled optical or specimen discretization error would require a separate allowance.

Across 24 synthetic phantoms and three noise realizations each, the learned reduced-field plates narrow mean voxel-wise bound width by about 21 %. On 48 neuronal windows, however, bounds span the full admissible range for every voxel at the tested settings. Better neuronal reconstruction in the main study therefore does not imply unique voxel intensities or uniquely identified connections. These negative neuronal results are retained as limits of the measurement-only analysis, not as optical-learning performance measures.

S10. Neuronal views and display selection

Illustrative unequal-peak example. The main neuronal figure uses SOM006_cell3_c0 with pretrained PnP at $C = 8$ and nominal dose. We screened the 60 references for interior columns with exactly two peaks, separated by at least three planes, and a weaker-to-stronger height ratio of 0.2–0.6. Peaks must meet the height and prominence thresholds used for layered-column evaluation. A supported region contains the center and at least four neighboring eligible columns with corresponding peak planes within one plane. This gives 685 eligible columns and 13 supported regions in six crops. An expanded screen with two supporting neighbors was also examined; the selected column meets the original, stricter rule. We inspected paired reconstruction outcomes and selected the displayed column for its lower W1 and better retention of the weaker component across all five subsets. This is an outcome-selected local illustration, not a representative or aggregate comparison.

Subset	Untrained	Ours	End-to-end
1	0.344	0.117	0.231
2	0.514	0.110	0.366
3	0.490	0.102	0.147
4	0.648	0.150	0.296
5	0.815	0.169	0.346
Mean	0.562	**0.130**	0.277

Table S23. Column W1 (μm) for the selected unequal-peak example across the five recorded direction subsets. Subset 1 is displayed in the main figure.

The column is $(x, y) = (35, 30)$, with reference peaks at planes 3 and 11 (zero-based) and a weaker-to-stronger height ratio of 0.59. The axial band spans rows 34–36. Table S23 and Fig. S5 give the five-subset errors and profiles, including raw intensity. Our column W1 is lower than both controls in every subset. Among the 13 original region centers, our and end-to-end W1 beat the untrained plate in all five subsets at 10 and 8 centers, respectively; each learned plate beats both others at one center. The larger region around the displayed column does not share its ordering: our and end-to-end mean W1 are 0.223 and $0.210\,\mu$m over its five columns, and 0.220 and $0.218\,\mu$m over the crop. The displayed volumes reproduce the recorded rows within 0.002 dB PSNR and $0.0003\,\mu$m crop W1.

Reference-only selection. Before opening reconstructions for figure selection, we computed reference features for all 60 crops. Eligible crops had at least 10 skeleton endpoints (the upper quartile) and a 10–90 % column-centroid range of at least $3.64\,\mu$m (the median), indicating several branches and substantial depth extent. Eight crops qualified. We ranked them by reference signal in columns with at least two axial peaks. The first was SOM004_cell2_x7; the companion was the highest-ranked eligible crop from another animal, SOM006_cell1_x7.

After selection, their W1-gain ranks among 60 crops were 52/45/23 and 45/46/43 for TV/pretrained/neuron-trained PnP, where rank 1 is the largest reduction. These are typical-to-lower-gain examples. We retain both alongside the main figure's unequal-peak illustration. Figures S6 and S7 show all three iterative methods for these crops. The full-cell panels supply two-photon context; only the marked crops generate simulated measurements.

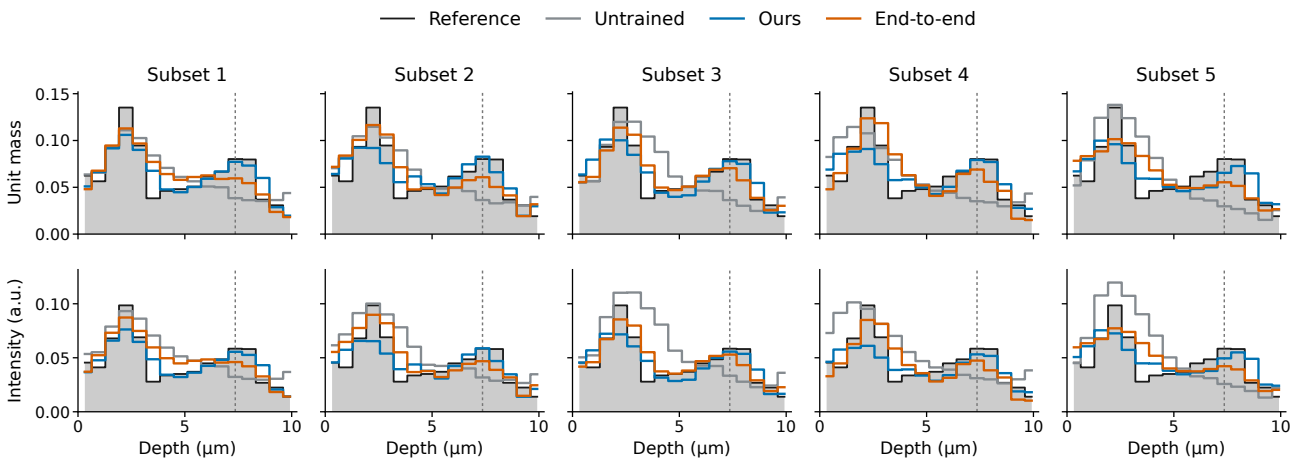


Figure S5. The selected main-figure column across all five direction subsets. Top: unit-mass axial profiles used for column W1. Bottom: raw intensities on one shared linear scale. Gray fills show the reference, and the dashed line marks its weaker peak. Our plate preserves this component's relative strength more closely across subsets; in Subset 1, it also underestimates the stronger peak most.

Views and scales. For the two reference-selected crops, the axial band maximizes reference signal in layered columns over three consecutive lateral rows ($1.92\,\mu\mathrm{m}$). Within it, the profile column has the largest reference signal among layered columns whose two highest peaks are at least three planes apart. This gives $(x, y) = (35, 44)$ in the first crop and $(42, 16)$ in the companion; the first lies three voxels from an edge and was retained unchanged. Their displayed volumes use noise draw 0. These columns favor the end-to-end plate more strongly than the crop-wide error; they illustrate local behavior, not a population ranking.

Depth-colored projections tint each plane with an approximately constant-lightness depth color at its linear intensity, then take the maximum independently in each color channel. Overlapping layers may mix colors or be dominated by the brighter layer, so axial bands, profiles and W1 maps accompany the projections. Fluorescence intensity images—bands, slabs, reference projections and two-photon contexts—use the same magenta map. Reconstruction panels use one linear range per crop, from zero to the 99.5th percentile of its reference maximum projection, with no gamma, smoothing or per-panel rescaling. W1 maps use a shared gray 0–1 $\mu\mathrm{m}$ scale: black at zero and white at or above $1\,\mu\mathrm{m}$. Unscored pixels outside the reference signal mask are also black. Rows and columns correspond to x and y throughout; axial depth increases downward.

Full-cell contexts are maximum projections of the original two-photon stacks, linearly clipped at their 99.8th percentile. They retain tissue coordinates ($0.25\,\mu\mathrm{m}$ lateral sampling), whereas reconstructed views use simulation coordinates. The boxed $32 \times 32 \times 32\,\mu\mathrm{m}^3$ tissue crops become $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$ simulated volumes. The reference-selected reconstructions reproduce the recorded rows within 0.018 dB PSNR and $0.001\,\mu\mathrm{m}$ W1. The following full views retain context and individual profiles.

A feature missed by both plates. A separate weak-feature illustration uses the earlier reference-selected companion band: the layered column with largest reference mass whose two highest peaks are at least three planes apart. Its selected column, $(x, y) = (18, 3)$, contains a weak shallow peak at plane 2 above a deeper peak at plane 9. The shallow peak is 24.5 % of the column maximum and carries 7.8 % of its signal in planes 1–3. None of the six untrained- and our-plate reconstructions has a matched shallow peak; that signal becomes a low plateau (Fig. S16). This limit coexists with lower average W1 and improved placement of brighter features.

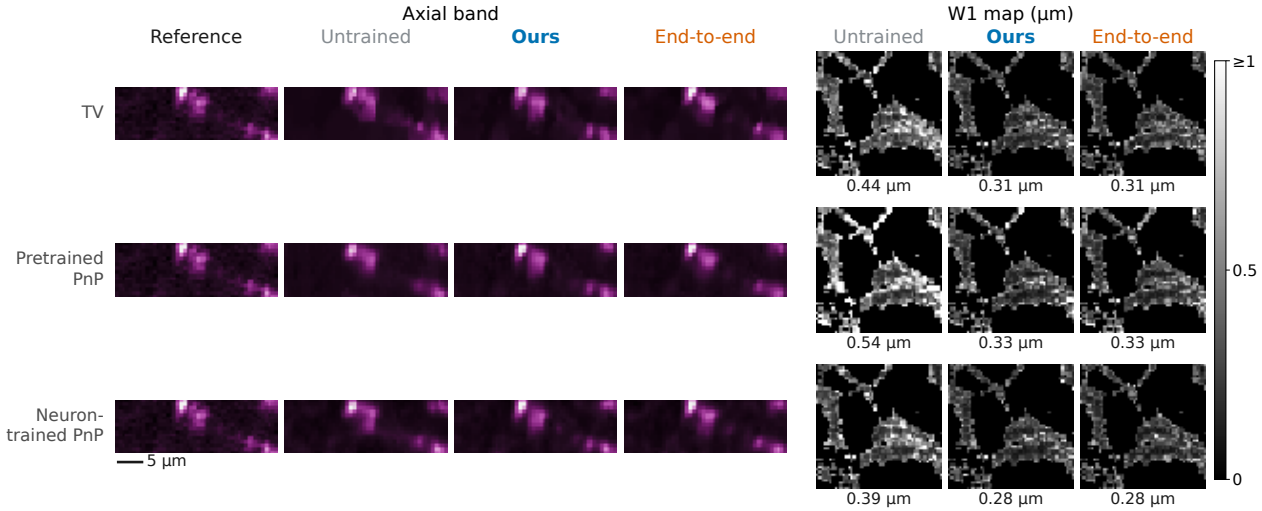


Figure S6. All three iterative methods for the first reference-selected crop. Left: matched axial bands for the reference, untrained, our and end-to-end plates. Right: column-W1 maps with draw-0 crop means below. Black also marks pixels outside the reference signal mask. The 5 μm bar applies to both axes of each band; maps span the full 30.72 μm lateral field.

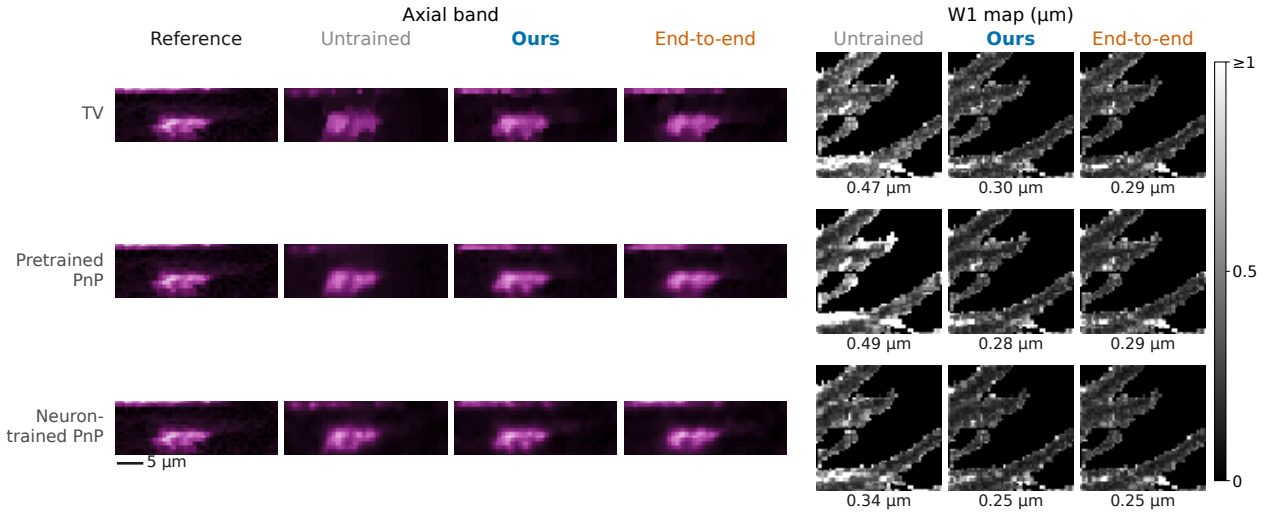


Figure S7. The companion crop from another animal, using the same band and error-map conventions as Fig. S6. Values are draw-0 crop W1 in μm; the bar is 5 μm.

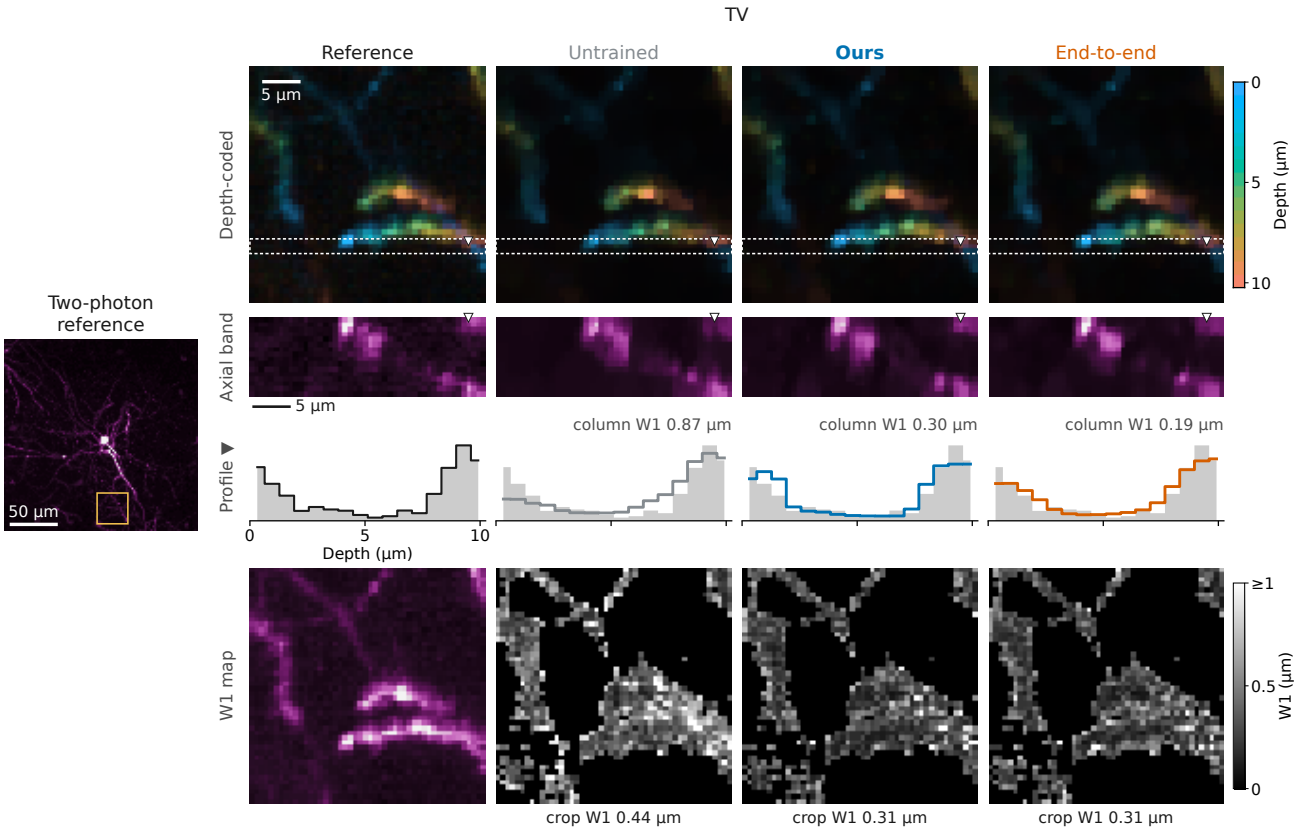


Figure S8. First crop with TV. Context, projection, band, profile and error-map conventions match the main neuronal figure. The marked profile is one reference-selected column; the means below the maps summarize all reference-signal columns for this draw.

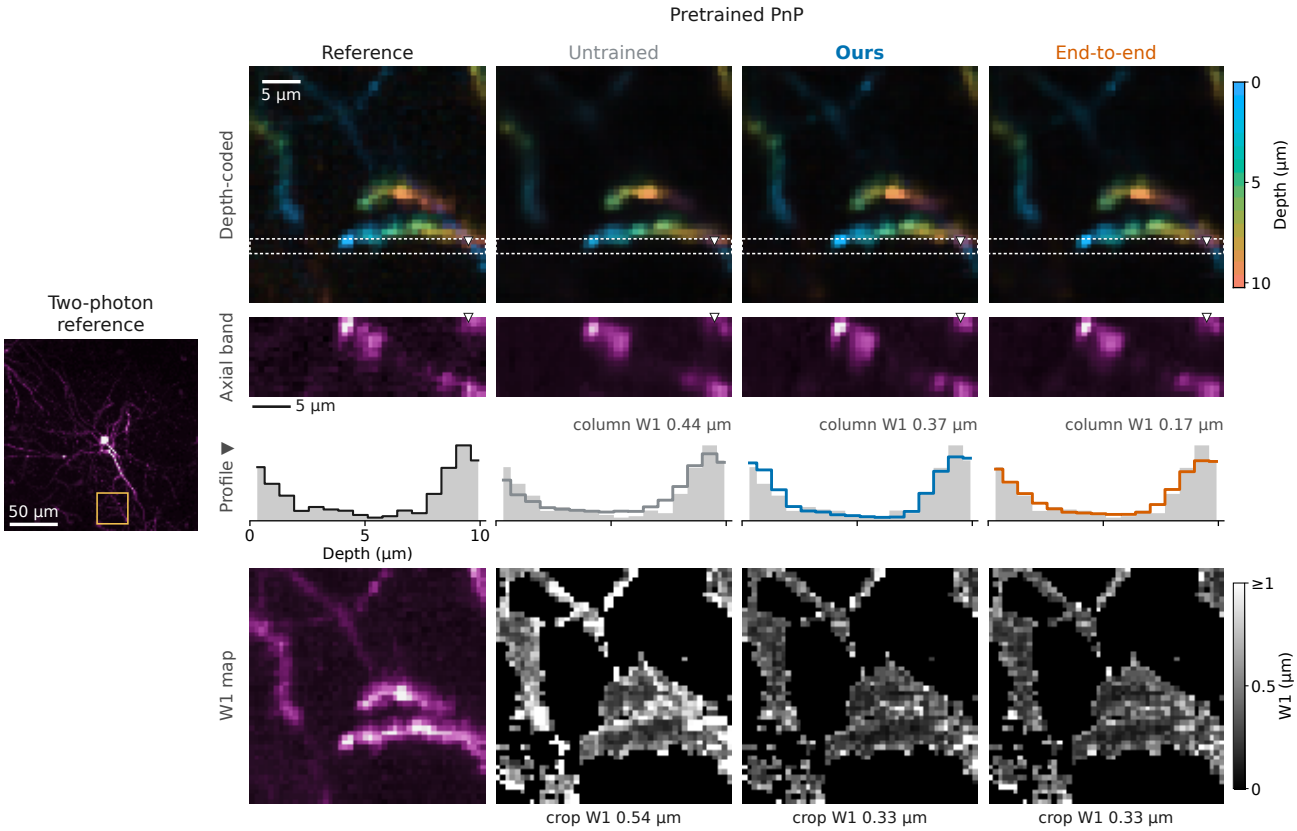


Figure S9. First reference-selected crop with pretrained PnP, retained from the earlier main figure. The marked column favors end-to-end more strongly than the crop mean.

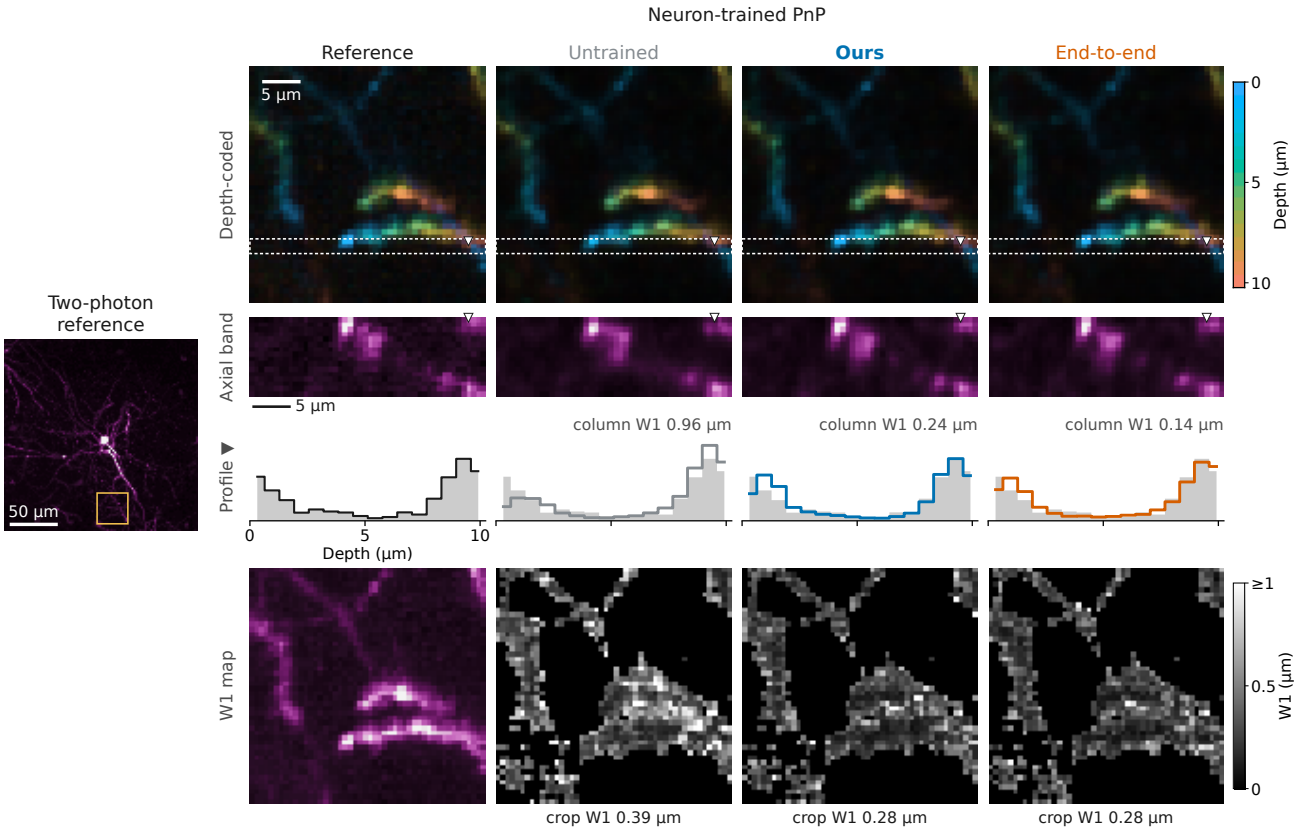


Figure S10. First crop with neuron-trained PnP. Context, projection, band, profile and error-map conventions match the main neuronal figure. The marked profile is one reference-selected column; the means below the maps summarize all reference-signal columns for this draw.

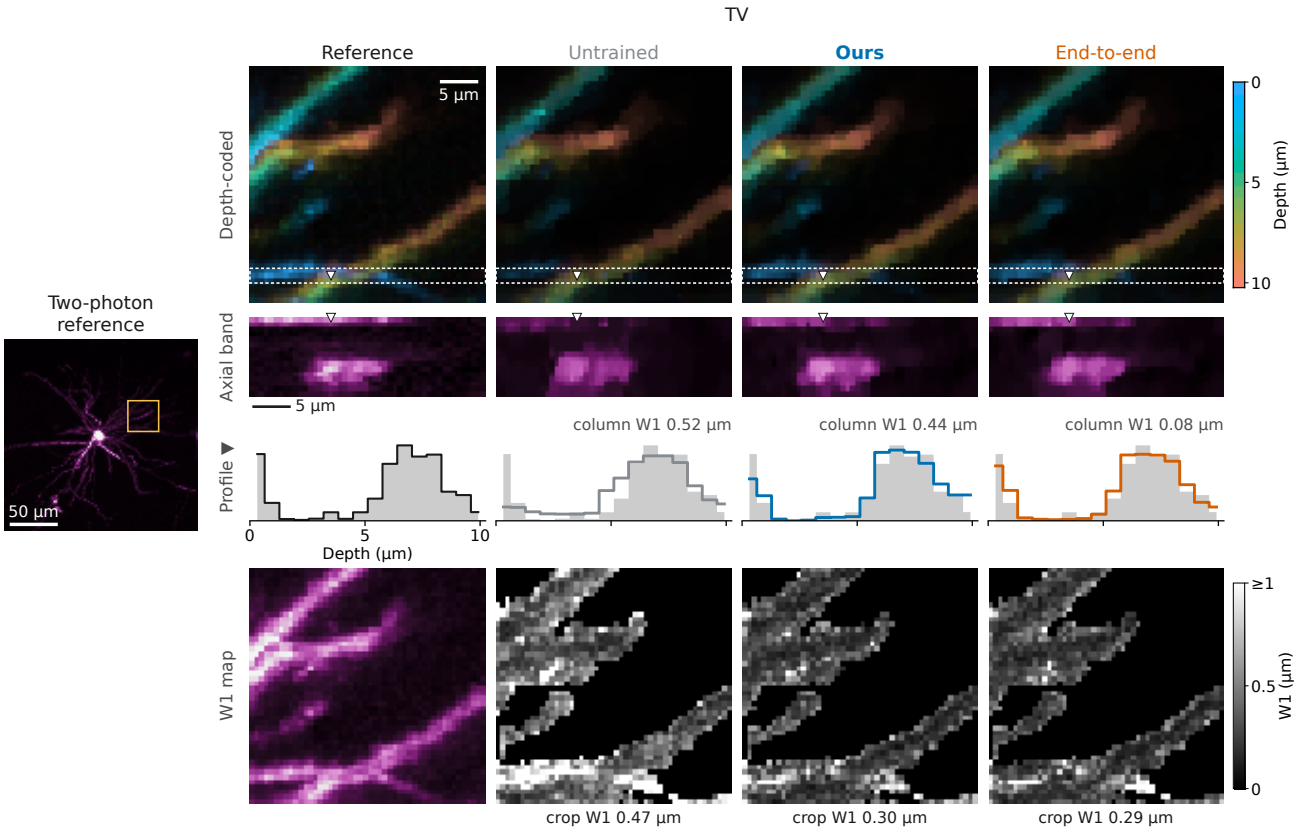


Figure S11. Companion crop with TV. Context, projection, band, profile and error-map conventions match the main neuronal figure. The marked profile is one reference-selected column; the means below the maps summarize all reference-signal columns for this draw.

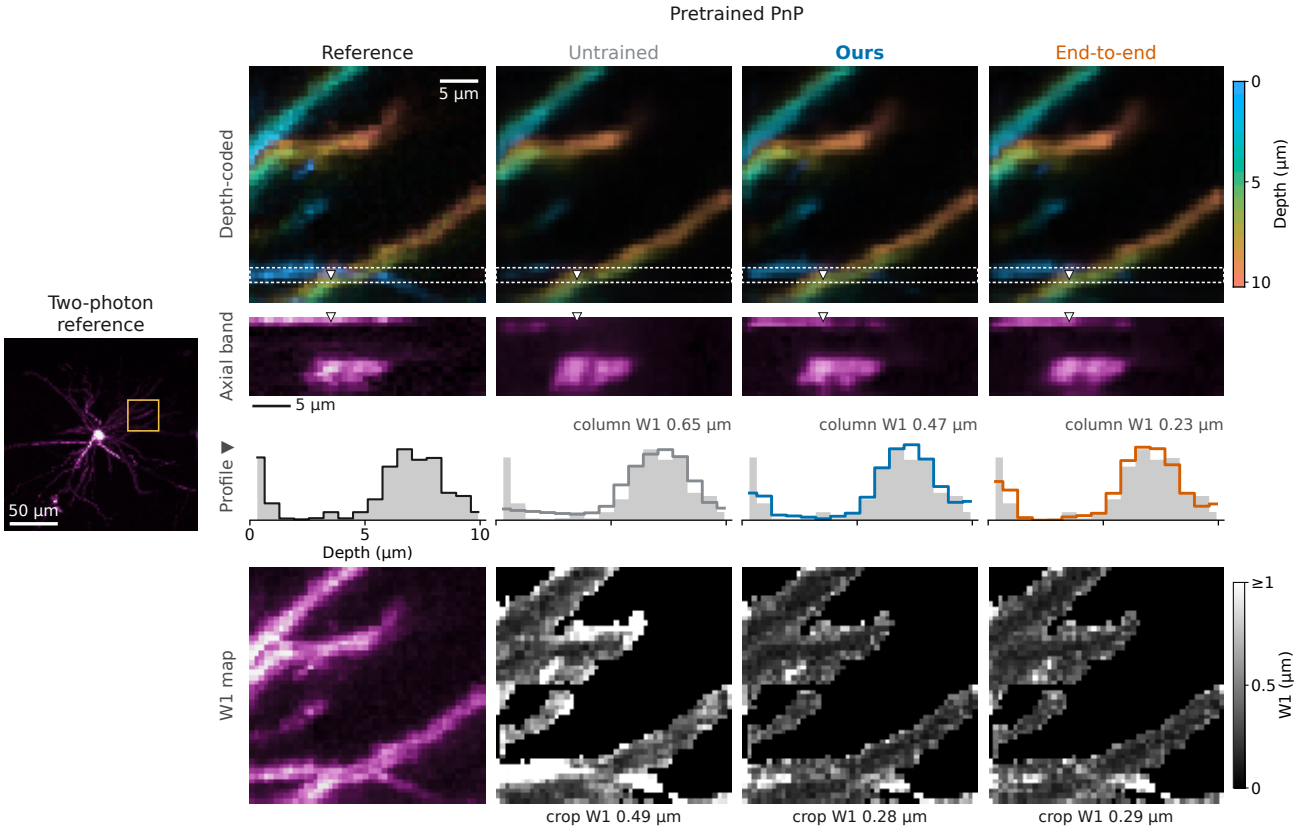


Figure S12. Companion crop with pretrained PnP. Context, projection, band, profile and error-map conventions match the main neuronal figure. The marked profile is one reference-selected column; the means below the maps summarize all reference-signal columns for this draw.

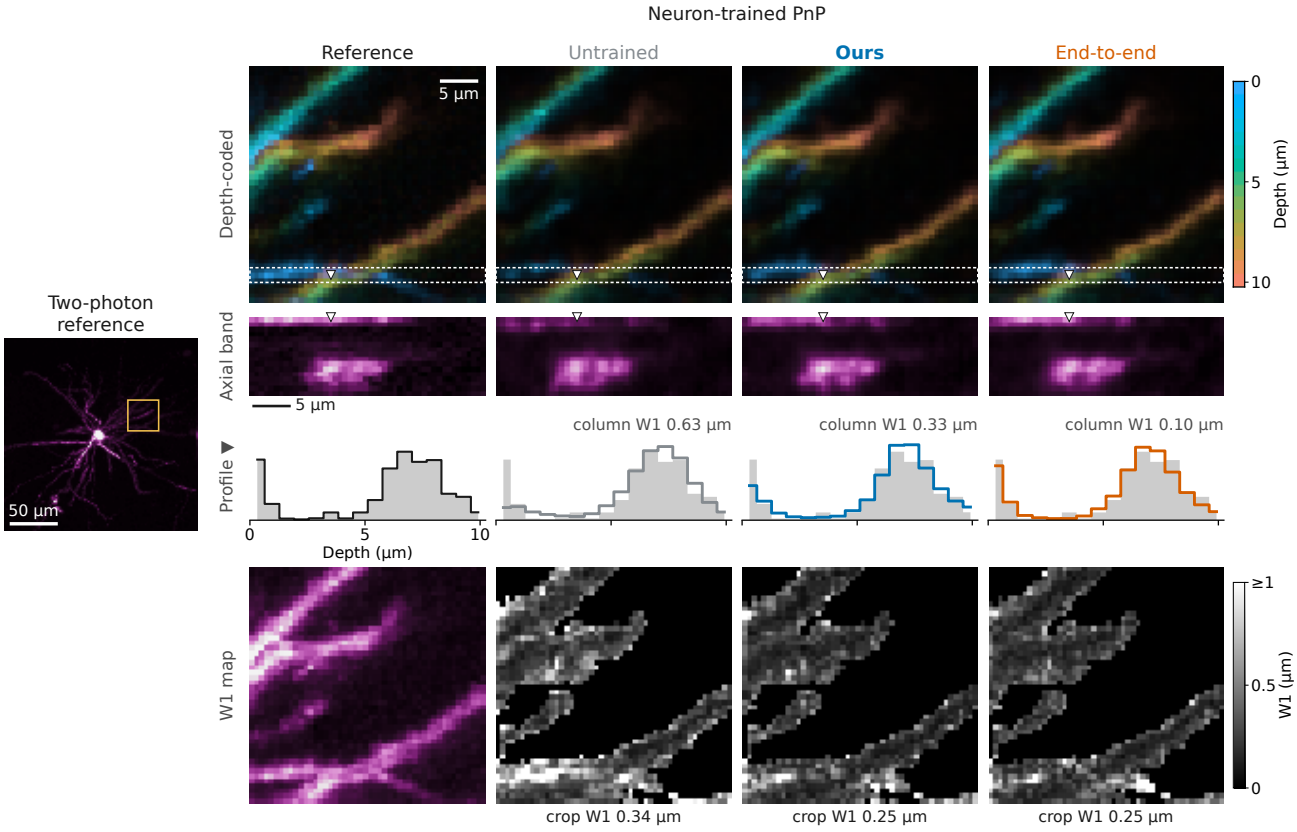


Figure S13. Companion crop with neuron-trained PnP. Context, projection, band, profile and error-map conventions match the main neuronal figure. The marked profile is one reference-selected column; the means below the maps summarize all reference-signal columns for this draw.

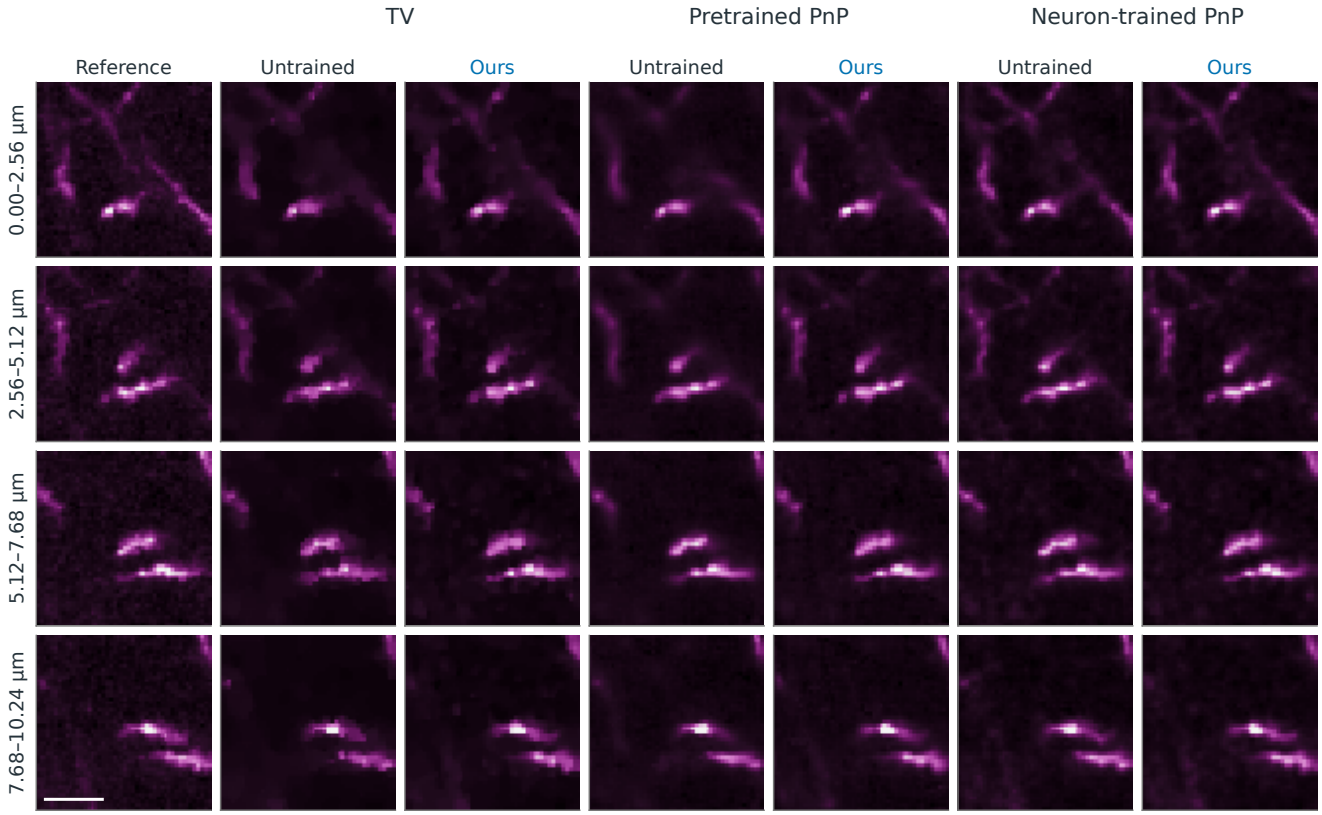


Figure S14. Depth slabs of the first neuronal example, with maximum projections over four consecutive depth planes ($2.56\,\mu$m per slab). Reference and all reconstructions share the same linear intensity range across slabs. Slab views expose signal spread into depths absent from the corresponding reference slab. These are qualitative placement comparisons, not branch-separation measurements. Bar: $10\,\mu$m.

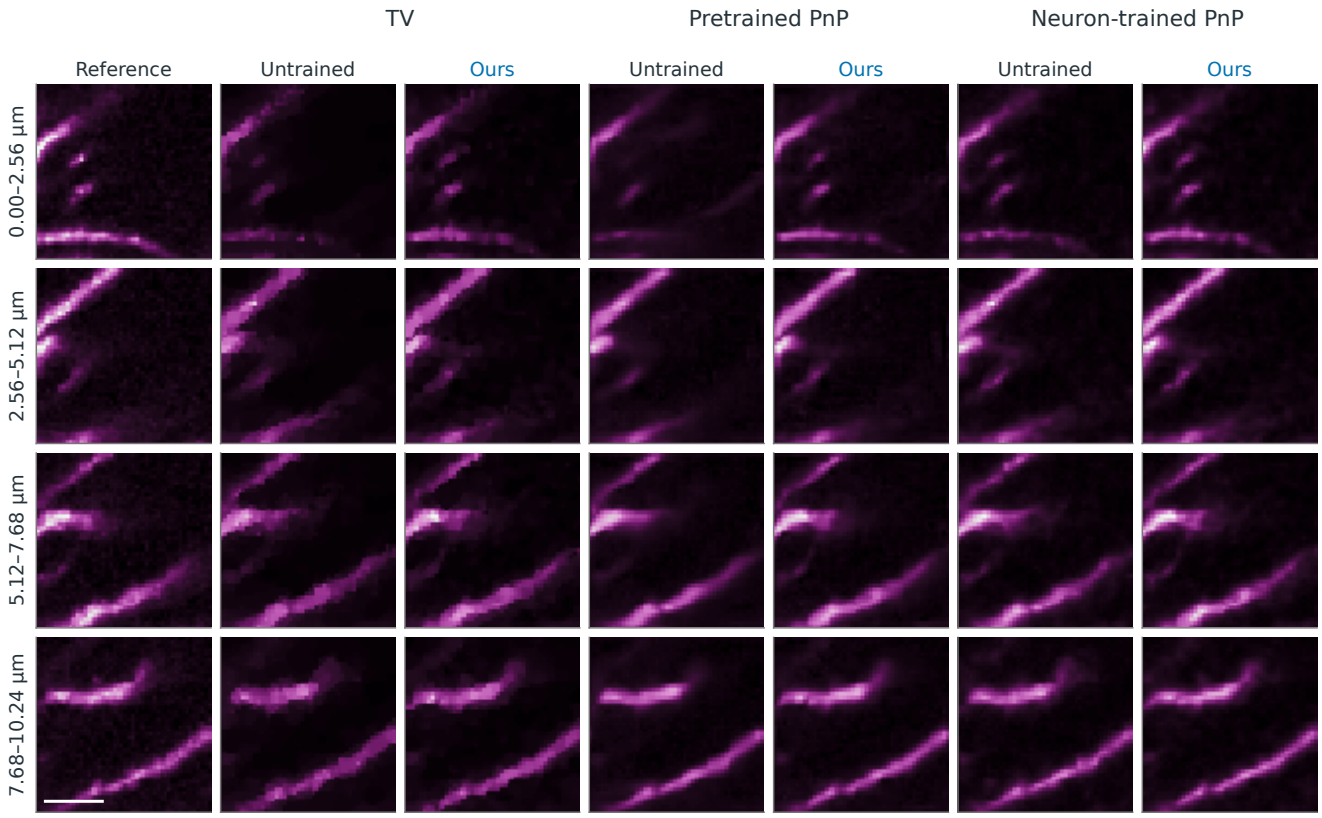


Figure S15. The companion crop from another animal, with the same depth-slab conventions. Stronger priors reduce diffuse signal for both plates; the optical difference is smaller under neuron-trained PnP. The full views retain features missed by either encoder.

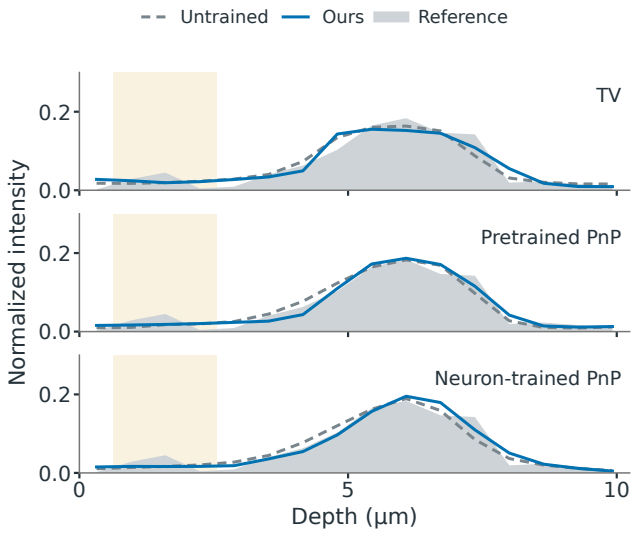


Figure S16. Reference-selected weak-feature column in the companion crop. Profiles are normalized to unit mass. Both plates miss the shallow reference peak under all three priors, despite the lower average W1 of our plate. This column was fixed from reference structure before inspecting reconstructions.

References

[1] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024. 16

[2] Thilo Balke, Fernando Davis, Cristina Garcia-Cardona, Soumendu Majee, Michael McCann, Luke Pfister, and Brendt Wohlberg. Scientific computational imaging code (SCICO). *Journal of Open Source Software*, 7(78):4722, 2022. 7, 8

[3] John J. Benedetto and Matthew Fickus. Finite normalized tight frames. *Advances in Computational Mathematics*, 18: 357–385, 2003. 3

[4] Julio Martin Duarte-Carvajalino and Guillermo Sapiro. Learning to sense sparse signals: Simultaneous sensing matrix and sparsifying dictionary optimization. *IEEE Trans. Image Process.*, 18(7):1395–1408, 2009. 3

[5] Michael Elad. Optimized projections for compressed sensing. *IEEE Trans. Signal Process.*, 55(12):5695–5702, 2007. 3

[6] Justin P. Haldar. On ambiguity in linear inverse problems: Entrywise bounds on nearly data-consistent solutions and entrywise condition numbers. *IEEE Transactions on Signal Processing*, 71:1083–1092, 2023. 17

[7] Q. Huangfu and J. A. J. Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018. 17

[8] A. Marian, F. Charrière, T. Colomb, F. Montfort, J. Kühn, P. Marquet, and C. Depeursinge. On the complex three-dimensional amplitude point spread function of lenses and microscope objectives: theoretical aspects, simulations and measurements by digital holography. *Journal of Microscopy*, 225(2):156–169, 2007. 1

[9] M. Milanese and A. Vicino. Optimal estimation theory for dynamic systems with set membership uncertainty. *Automatica*, 27(6):997–1009, 1991. 17

[10] B. Richards and E. Wolf. Electromagnetic diffraction in optical systems, II. structure of the image field in an aplanatic system. *Proceedings of the Royal Society of London. Series A*, 253(1274):358–379, 1959. 1

[11] Yoav Shechtman, Steffen J. Sahl, Adam S. Backer, and W. E. Moerner. Optimal point spread function design for 3D imaging. *Physical Review Letters*, 113(13):133902, 2014. 4

[12] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. clDice – a novel topology-preserving loss function for tubular structure segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16555–16564, 2021. 12

[13] Navodini Wijethilake, Mithunjha Anandakumar, Cheng Zheng, Peter T. C. So, Murat Yildirim, and Dushan N. Wadduwage. DEEP-squared: deep learning powered De-scattering with Excitation Patterning. *Light: Science & Applications*, 12:228, 2023. 8

[14] Kyrollos Yanny, Nick Antipa, William Liberti, Sam Dehaeck, Kristina Monakhova, Fanglin Linda Liu, Konlin Shen, Ren Ng, and Laura Waller. Miniscope3D: optimized single-shot miniature 3D fluorescence microscopy. *Light: Science & Applications*, 9:171, 2020. 3

[15] Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7):3142–3155, 2017. 7